

Dynamic Modelling and Control of a Grid-Connected PEM Electrolyser Using PID and MPC

Kasper Bjarke Larson
Student number: 202108752

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Supervisor: Ali Masoudian
Examiner: Mohammad Hassan Khooban

Abstract

As the energy sector moves toward green hydrogen, proton exchange membrane (PEM) electrolyzers have emerged as strong candidates for direct coupling with intermittent renewable sources, due to their fast transient response and robustness to varying operating conditions. However, their modelling and control under realistic grid disturbances are lacking in research, and comparisons of conventional and modern control strategies are limited. This project investigates the dynamic modelling and control of a grid-connected PEM electrolyser, comparing a PID controller against a Model Predictive Controller (MPC).

A complete system was developed in Simulink at component level, consisting of a three-phase source, a three-phase full-bridge rectifier, a three-phase interleaved buck converter, and a dynamic equivalent circuit model of a 50-cell PEM stack rated at $100kW$. A linearised transfer function capturing the electrolyser dynamics was derived to drive the MPC. Both controllers were used to regulate electrolyser power draw and were evaluated regarding noise and ripple suppression, harmonic distortion, interleaved-switch failure, stepped target values, and phase-to-ground grid faults.

The results show that both controllers behave similarly on a macro scale and share identical rise times, but differ in the short-term. The MPC achieved lower overshoot (0.251% vs 10.36%) and faster settling (11.975 ms vs 29.115 ms), with superior suppression of switching dynamics, at the cost of a small steady-state error that the PID eliminated entirely. Both handled grid faults well due to significant voltage headroom. The findings confirm the MPC as a viable and robust control method for PEM electrolyzers, while highlighting modelling limitations and possibilities for future multi-variable control.

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Chapter 1

Introduction

As the world moves toward net-zero carbon emissions, the transition from fossil fuels to renewable energy sources (RESs) has become a central challenge for modern society, and while the transition is already underway and is expected to disrupt key sectors of industry [24], industry still heavily relies on fossil fuels. One area which is almost completely dominated by coal and gas is hydrogen production, with methane reforming or coal gasification being the main methods of generation, and producing 65% and 25% of the world's hydrogen respectively. [17]. Hydrogen is mainly used in oil refinement, ammonia & methanol production for fertilizer, and steel manufacturing[17]. The sustainable alternative to the methods currently used is known as electrolysis, where electricity is used to split water into hydrogen and oxygen with high purity. Currently a total of 96% of hydrogen used in industry is generated from fossil fuels, while only 4% is generated through use of electrolysis, and only 1% of worldwide hydrogen is produced from electrolysis using renewable sources[23, 17]. Electrolysis has the potential to save millions of tonnes of CO_2 every year depending on how the electricity is generated[31]. Electrolysers using power from a mix of renewables and fossil fuels are estimated to emit 3.94 kg to 34.85 kg CO_2 -eq/kg H_2 depending on the exact mix [19]. Interestingly, this is potentially higher than the equivalent estimated emissions of generating hydrogen using SMR, at around 11.57 kg CO_2 -EQ/kg H_2 . This makes it clear that if electrolysis is to supersede conventional fossil fuel methods, the power needs to be derived from renewable sources, not just primarily, but exclusively[5].

Traditional alkaline electrolyzers (AELs) are easily scalable and have relatively low initial costs, however recent developments have introduced alternatives such as the Proton Exchange Membrane, which feature much higher hydrogen purity, as well as faster transient times and are more robust to changes in operation parameters such as supply voltage. This makes them potentially viable candidates for direct coupling with intermittent renewable sources such as wind-turbines & solar panels and a possible front liner for the switch from oil and gas powered grey hydrogen, to green hydrogen from renewable energy.

Despite the growing relevance of PEM electrolyzers as candidates for direct coupling with intermittent renewable sources, a number of challenges in their modelling and control remain insufficiently addressed in the literature. Existing electrolyser models are often either high-complexity electrochemical models with significant computational cost, and thus unsuitable for real-time control design, or they simplified static models that capture only steady-state polarisation behaviour and omit the electrical dynamics necessary for transient analysis [4].

Furthermore, while PID controllers are widely used for power converter control, they operate solely on an error signal and have no mechanism to incorporate system constraints or secondary objectives such as temperature limits or hydrogen production targets into the control decision. This shows a need to explore other potential control methods and examine which might be better suited for a task such as controlling a PEM electrolyser. One of such methods which is explored in this project is the Model Predictive Controller

(MPC) The MPC generates a control signal by optimising a cost function instead of simply relying on a single error variable, making it well suited to handle multi-variable systems such as electrolyzers. Systematic comparisons of PID and MPC for PEM electrolyser converter control under realistic disturbance scenarios remain limited. MPC has been applied to PEM electrolyzers for temperature regulation and grid frequency support as well as current control, but has not to the knowledge of the author, been explored and compared to PID under faults and failure scenarios [28, 25]. This project thus aims to investigate the control and mathematical modelling of PEM electrolyzers and simulate the complex interactions between source and electrolyser in realistic scenarios which a grid connected electrolyser might face (Faults, power intermittency, harmonic noise and switch failure).

Water electrolyzers for hydrogen production exist on many scales from large industrial megawatt installations to plants using a few hundred watts [6]. For industrial applications, the electrolyser is simply a single part of a larger system which handled transportation of gasses and liquids, power delivery for both electrolyser surrounding devices, and cooling systems keeping the electrolyser from overheating [38]. Electrolysers are DC devices, meaning a connection to the grid requires power conversion to get from AC to DC, as well as a step down in voltage due to the low voltage operation of electrolyzers. This conversion and step-down requires fine control to keep the electrolyser within operating specifications, as well as being able to control the amount of generated hydrogen, this is especially important when connected to intermittent renewable sources where voltage and power is constantly changing [44].

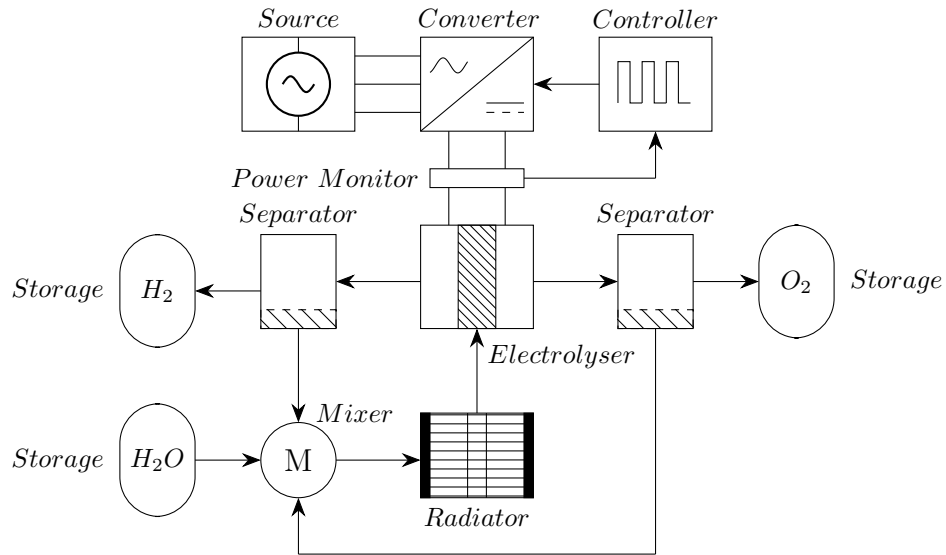


Figure 1.1: Overall schematic model of a PEM electrolyser system including source, converter, fluid interfaces, cooling system and control system

The water for the electrolyser is pumped into the device at a constant controlled rate. For AEL, the water is an alkaline solution; for PEM it is deionized.[2]. In the case of PEM electrolyzers, the water is run through multiple purification and demineralisation steps before entering the electrolyser itself[36], for AELs, the mix is made ahead of time and recirculated while monitored such that as the water is converted the additional water is added to make sure the concentration stays within a specified range. The electrolyte itself remains in the system and it not consumed by the reaction[2].

Due to the high heat and pressure within the stack, the generated gases contain traces of water vapour that must be removed before the gas can be stored. Gas-liquid separators are connected directly to the output fluid pipes to remove any excess water, which is then pumped back into the system to prevent needless water-waste. Finally the gases are compressed and stored or distributed.[11, 14]

During operating, the system generates excess heat as well. Electrolysers function best at around

60-80°C, so to cool the device, a separate liquid cooling loop is installed, using control logic to track the stack temperature and adjust cooling as needed. If the temperature cannot be controlled by cooling alone, the applied current is reduced to decrease the heat generation rate of the system. [30, 42]

1.1 Alkaline

The alkaline electrolyser (AEL) is the backbone of all modern hydrogen electrolysis and been in use since 1789[12]. AELs are relatively cheap up front compared to their PEM counterpart, and are very scalable on an industrial level. They are made up of individual electrolytic cells that combine into larger stacks. The AEL works by passing an electrical current through an electrolyte solution. When exposed to an electrical current, the solution will react at the electrodes, forming oxygen gas at the anode and hydrogen at the cathode. Between the two electrodes is a porous diaphragm, traditionally made from asbestos or potassium titanite, more commonly now made from polypropylene or polyphenylene. This diaphragm keeps gasses from passing between the anode and cathode region[37]. Preventing mixing of the two gasses is vital for electrolyzers for efficiency and safety reasons.

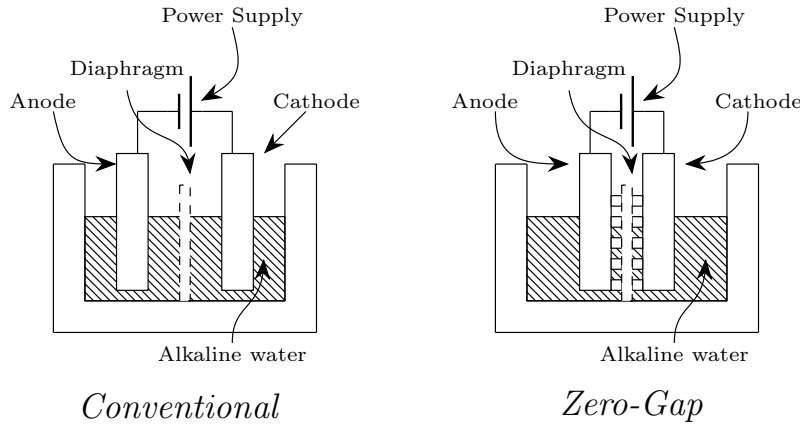


Figure 1.2: Comparison between sectional views of conventional (Left) and zero-gap (Right) alkaline electrolyzers. Conventional setup shows electrodes spaced apart from the diaphragm while the zero-gap setup feature gap-less and porous electrodes as to minimize the ion path and increase efficiency

The original design of the AEL features simply the two electrodes submerged in an electrolyte the diaphragm spaced between them. However, modern AELs use a zero-gap design where the electrodes, instead of being spaced apart from the diaphragm, are moved up against it. This is advantageous, as the distance between the electrodes in the conventional electrolyser cause hydroxide ions to travel unnecessary distances, resulting in higher ohmic losses.[26] Moving the electrodes close together results in shorter travel and higher efficiency. It does however come with a new challenge, as having the electrodes flush against the diaphragm starves both regions of alkaline solution, thus decreasing generation rate. The structure of the electrodes are therefore made porous to prevent this starvation, which not only allows the electrolyte solution and gasses to pass between the electrodes and digraph, but also yields a higher surface area for reactions to take place, lowering the required voltage and enabling higher levels of generation for the same power cost[27, 13]. As cells are combined into stacks they are connected electrically in series, and hydrologically in parallel, such that all cells receive equal amounts of the alkaline solution, while carrying the same current[29, 40].

The water entering the AEL isn't purified as mentioned during the introduction, but a solution consisting of water and an electrolyte. While the reaction is still possible using pure water, the addition of an electrolyte makes the reaction much more effective, and thus, allows for more hydrogen to be generated per unit time. The electrolyte of choice is usually either Potassium Hydroxide (KOH) or Sodium Hydroxide ($NaOH$).

Of the two, Potassium Hydroxide is the more efficient choice, and tends to be chosen more frequently in modern industrial plants.

While the diaphragm generally aims to prevent transport of gasses between the anode and cathode region, it doesn't completely prevent it. Large amounts of either gas in the opposite chamber will in the best case cause lower overall efficiency of the system and in the worst case be a safety concern. The mix of hydrogen and oxygen gasses, known as oxyhydrogen can be highly explosive in certain ratios. For AELs operating at low pressure around 80°C , the explosive ratio limit is in the interval of roughly 4% – 94% hydrogen to oxygen[20]. The danger is much higher for hydrogen entering the anode, as a flammable ratio not only occurs at lower concentrations (4%), but hydrogen also diffuses faster through the diaphragm than oxygen [32]. Since oxygen has more difficulty moving through the diaphragm, and requires a higher concentration in hydrogen to become flammable (6%), oxygen presence in the cathode is generally monitored, but mostly contributes to lower efficiencies and excess heat generation.[33, 12, 35]

Chemical Reaction

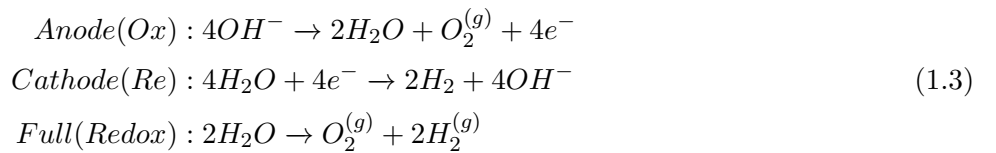
The reaction driving electrolysis is known as a redox reaction, which means that it's the product of two half-reactions, an (ox)idation and (red)uction reaction. For AELs, the electrolyte is pumped in at the cathode side, which, together with the free electrons from the electrical circuit, reduces into hydrogen gas and hydroxide (Eq. 1.1)



The hydrogen gas is pumped out of the electrolyser while the hydroxide, being negatively charged anions, moves through the porous diaphragm to the anode, where it upon contact causes an oxidation reaction by giving the free electrons of the hydroxide to the circuit, becoming water and oxygen. 1.2



Together, these two half-reactions make up the redox of the AEL, which shows that for each two molecules of water, one molecule of oxygen and two molecules of hydrogen is created. Water is thus converted completely into the two gasses without any waste-product and the electrolyte remains in the system.



The two reactions don't happen at the same rate, or have the same energy requirements, the oxidation reaction is far slower and energy-hungry than the reduction, bottlenecking the system[41, 15]. This will become relevant when modelling the system in section 2.1.2

1.2 Proton Exchange Membrane

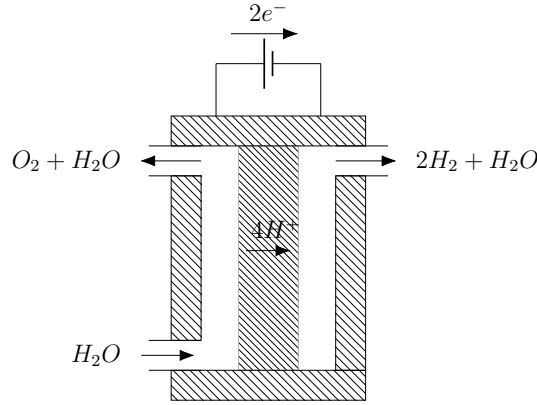


Figure 1.3: Cross-sectional view of PEM electrolyser including membrane, circuit and movement of reacting parts. Bottom arrow shows intake of water (H_2O) from separate system, upper left is the exiting of generated oxygen (O_2) and trace water, membrane shows travel of free protons ($4H^+$) into cathode chamber where recombination occurs and generates hydrogen which is seen exiting in the upper left along with trace water.

In the AEL, the electrolyte is the liquid solution of water mixed with an alkaline chemical such as KOH or NaOH as explored in section 1.1. The main difference between the AEL and the PEM, is the inclusion of a non-permeable diaphragm which functions as the electrolyte called a membrane. Originally made from sulphonated polystyrene [34], now usually from Nafion. When submerged in water, the acidity of the membrane allows for free hydrogen ions to pass, again, hindering the migration of electrons, which have to move through the circuit to get the opposite region and cause the reduction reaction[35]. Due to the membrane being an electrolyte in itself, the liquid used in the PEM is deionized water. While gas-crossover is lower for PEM than AEL, a small amount of hydrogen and oxygen permeation through the membrane is unavoidable in reality [1].

The existence of the membrane mixed with the fact that water is pumped in through the anode instead of the cathode like with the AEL, the chemical reactions that make up the redox reaction are slightly different, (Eq. 1.4, 1.5). The overall redox reaction however is identical. For both AEL and PEM, two molecules of water are consumed to produce two molecules of hydrogen and one molecule of oxygen, with no by-products (Eq. 1.6).

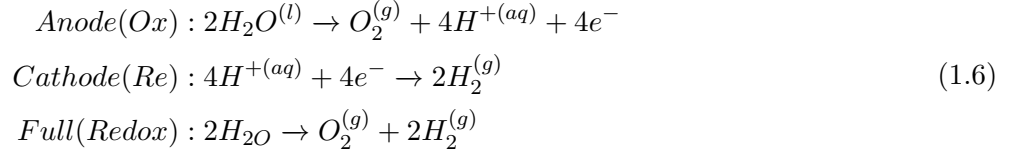
Chemical Reaction

At the side of the anode, the water that comes into contact with the electrode splits into oxygen, free protons and free electrons. The free electrons travel through the circuit and into the cathode chamber, while the free protons pass through the membrane due to the electric field generated by the two electrodes, and end up in the cathode chamber as well. The oxygen generated is pumped out and separated from any water vapour that has accompanied it.



On the cathode side, the free protons and free electrons combine at the electrode through reduction, and become pure hydrogen gas, which, in the same manner as with the oxygen, is pumped out and separated from any water that might be present.





While water is only pumped into the anode chamber, it is beneficial for the PEM electrolyser to have small amounts of water present in the cathode chamber as well, as the membrane requires hydration to maintain its proton conductivity [35]. This occurs naturally through two mechanisms; permeation due to pressure differences across the membrane, and electro-osmotic drag, where migrating protons carry water molecules through the membrane alongside them [22]. The water content on the cathode side is monitored to keep levels below a threshold, as excess water would flood the cathode and impede hydrogen output.

The following project only makes use of the Proton Exchange Membrane electrolyser as it is an emerging technology and worth researching to find out if it might supersede the AEL or even conventional grey hydrogen methods. Due to how similar the AEL and PEM is in theory, both are explained to let the reader understand the fundamentals of electrolysis, as well as give an understanding of the overall differences between the two technologies.

Chapter 2

Modelling

In order to model the system, a set of simplifying assumptions are made to keep the scope of the project realistic. The cell temperature and pressure is assumed constant throughout operation and are not modelled. Concentration losses are neglected, and the electrolyser stack is treated as spatially uniform, meaning all cells in the stack are assumed to behave identically. Finally, all sensors are assumed ideal, introducing no noise or measurement delay into the control loop. With these assumptions established, each section of the system is studied thoroughly and replicated in Simulink such that the simulated system represents the physical counterpart as closely as possible. If control is possible in this setting, it is thought to be likely that it would also be for an actual plant. Figure 2.1 shows the full PEM system with a three-phase generator feeding into a rectifier to convert the alternating current to direct current. From there the voltage is stepped down to a level suitable for the electrolyser through a buck converter, and finally the power is fed into the PEM electrolyser itself to generate hydrogen and oxygen. To keep the power draw of the electrolyser at a target value, the controller reads sensor data from the stack regarding current and power levels. The sensor signal is compared to a target value and used to generate a control signal for the buck converter.

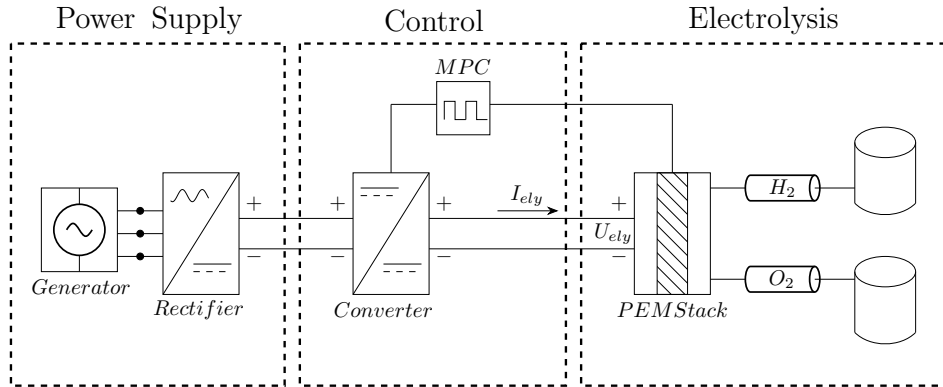


Figure 2.1: Full system overview showing the general movement of power and control circuitry

2.1 PEM model

The first part of the system that will be studied electrically here is the electrolyser. Section 1.2 already went into detail regarding the chemical and overall theory of the electrolyser. This section will thus only concern itself with the electrical steady state and dynamic response of the PEM electrolyser.

At steady state, the electrolyser has a non-linear relation between voltage and current, the resulting graph from plotting this relationship is called the polarisation curve, and can be expressed mathematically through as a sum of overpotentials of the electrolyser which stem from chemical interactions like the anode/cathode

activation (U_{act}), reversible voltage (U_{rev}), ohmic membrane voltage (U_{ohm}) and the concentration voltage (U_{con}). Each cell has these individual contributions to the overall voltage-drop, which are simply added together in a stack since, for this reason, assuming each cell is identical, a stack can simply be defined as a single cell multiplied by the stack count.

$$U_{cell} = U_{rev} + U_{act} + U_{ohm} + U_{con} \quad (2.1)$$

Since each cell in the stack is assumed identical, the total stack voltage follows directly as:

$$U_{stack} = N_{cells} \cdot U_{cell} \quad (2.2)$$

where in the case of this project, $N_{cells} = 50$.

The different contributions dominate at different current densities. The reversible voltage represents the minimum thermodynamic requirement for any electrolysis to occur. At low current densities, the activation voltage dominates, reflecting the energy barrier for the electrode reactions. In the mid-range operating region, ohmic losses become the primary contribution, increasing linearly with current. At very high current densities, the concentration voltage dominates as mass transport limitations prevent sufficient water from reaching the electrodes. Figure 2.2 shows each of these contributions individually as well as the combined polarisation curve.

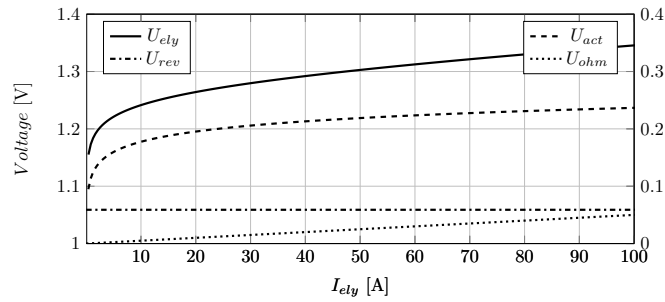


Figure 2.2: PEM cell voltage plot (Polarisation curve) separated into contributions. Leftmost side features y axis of reversible and cell voltage. Rightmost side features y axis for activation and ohmic membrane voltage. Concentration overpotential has been omitted due to project scope as will be explained in 2.1.4

2.1.1 Reversible Voltage

Chemically, the reversible voltage is the required voltage across a PEM cell for the electrolysis reaction to occur. For any potential below this threshold, the PEM electrolyser will act as an open circuit, and no current will pass through the cell.

At voltage potentials above this threshold, but below the thermoneutral voltage (the cell voltage at which heat generation is exactly zero, discussed in detail in Section 2.1.5) the electrolyser will draw enough power from the supply as to be able to get the remaining required power from the surrounding heat, causing a slight cooling effect while generating very low amounts of hydrogen.

$$U_{rev} = 1.5184 - 1.5421 \cdot 10^{-3} \cdot T + 9.523 \cdot 10^{-5} \cdot T \cdot \ln(T) + 9.84 \cdot 10^{-8} \cdot T^2 \quad (2.3)$$

The reversible voltage depends on the temperature and pressure of the system. However in this system, the pressure part has been omitted for simplification reasons as was stated in the beginning of chapter 2. Most of the overpotentials of the electrolyser depend in some manner on the temperature of the stack, and thus, the operating point of the electrolyser can vary greatly with changes in temperature as seen in figure 2.3.

2.1.2 Activation Overpotential

The activation overpotential is the contribution to U_{cell} stemming from the electrodes of the system. As the current density of the electrodes increases, the required voltage follows in a logarithmic manner. The governing equation is the Table equation [9], a simplification of the Butler-Volker equation. The equation depends on the current density at the electrode (i), the exchange current density (i_0), and the temperature of the system (T). Constants F and R are the Faraday and universal gas constant respectively, $\alpha = 0.5$ is the charge transfer coefficient, and $Z = 2$ is the stoichiometric number of electrons transferred in the global half-reaction[16].

$$U_{act} = \frac{R \cdot T}{\alpha Z \cdot F} \cdot \ln \left(\frac{i}{i_0} \right) \quad (2.4)$$

Since the anode and cathode each contribute their own activation overpotential, a fully general model separates the two. However, as the exchange current density at the anode is several orders of magnitude lower than at the cathode, around 10^{-7} A/cm² compared to 10^{-3} A/cm² at 80 °C [16], the anode contribution dominates by a lot, and thus the cathode term can be neglected without much loss in accuracy.

2.1.3 Ohmic Overpotential

The ohmic overpotential U_{ohm} is caused by losses as a result of proton conduction through the membrane. As the name suggests the relationship between voltage and current for this overpotential is linear. The resistance of the membrane is calculated as the thickness of the material divided by the proton conductivity area (EQ. 2.5)

$$R_{mem} = \frac{t_{mem}}{\sigma A} \quad (2.5)$$

$$\sigma = (0.005139\lambda - 0.00326) \cdot e^{1268(\frac{1}{303} - \frac{1}{T})} \quad (2.6)$$

$$U_{ohm} = R_{mem} \cdot I \quad (2.7)$$

2.1.4 Concentration voltage

Concentration losses have been omitted in the same manner as the pressure term in the reversible voltage (Sec. 2.1.1). At the operating point of this system (1000 A across the 50-cell stack at 100 kW as will be shown in section 4, which corresponds to a cell voltage of approximately 2 V per cell) concentration losses may not be entirely negligible. However, as the primary goal of this project is to investigate control methods for PEM electrolyzers rather than to produce a high-fidelity electrochemical model, omitting this term is accepted as a modelling simplification. The resulting polarisation curve will overestimate performance at high current densities, and this should be kept in mind when interpreting results near the upper operating limit.

2.1.5 Temperature dependence and efficiency

As heat is produced through the chemical reactions inside the electrolyser, the temperature of the system increases, which in turn changes the dynamics of the electrolyser itself as all the overpotentials making up the cell voltage is dependent on the temperature. Generally for electrolyzers, an increase in temperature corresponds to a lower cell voltage, which allows for higher current levels for the same voltage, increasing production. For this reason, in the case of constant current operation, the electrolyser will become more efficient as it gets up to temperature.

The generated heat can be expressed as an integral of the net total thermal power, where power levels above those derived from the thermoneutral voltage result in heating, and power levels below result in a cooling effect, as described in Section 2.1.1. The variables are the thermal resistance R_t ($\frac{K}{W}$), the thermal capacitance C_h ($\frac{J}{K}$), and the ambient temperature T_{amb} (K). All temperatures in the thermal model are expressed in Kelvin.[39, 43].

$$T = \int \left((Q_{gen} - Q_{loss}) \frac{1}{C_h} \right) \quad (2.8)$$

$$Q_{gen} = P_{ely} - I_{ely} \cdot U_{tn} \quad (2.9)$$

$$Q_{loss} = \frac{T - T_{amb}}{R_t} \quad (2.10)$$

The last variable used in modelling the thermals of the PEM electrolyser is the thermoneutral voltage. The thermoneutral voltage is the cell voltage level at which the temperature stays constant, and thus, any value above it result in heating. It is also what defines the thermal efficiency of the electrolyser. Since values above it causes losses to heat, the most thermally efficient operating point is a cell voltage equal to that of the thermoneutral voltage, where losses to heating in the chemical parts of the system is essentially zero. However, as the thermoneutral voltage is quite low (1.48V at 25°C). Operating at this point will lead to low hydrogen generation and will thus be economically inefficient.

$$U_{tn} = 1.4045 + 1.5784 \cdot 10^{-4} \cdot T + 3.8037 \cdot 10^{-7} \cdot T^2 \quad (2.11)$$

Most electrolyzers operate at their maximum economic efficiency at around 50 – 80°C depending on current electricity prices with 80°C being the upper limit for most electrolyzers. At these temperatures, the electrolyser operates at its highest output without exceeding temperature bounds [8]

$$\eta_{ely} = \frac{U_{tn}}{U_{ely}} \quad (2.12)$$

It is clear from figure 2.3 that, the higher the temperature, the lower the voltage requirement for a target current level. The heating of an electrolyser stack, and thus the change in voltage resulting from it, is a very slow dynamic relative to the electrical and control dynamics of interest in this project[7]. For this reason, temperature is treated as a fixed parameter in the model, as stated in the assumptions. This simplification means the controller is designed and validated at a single operating temperature; in practice, the polarisation curve will shift as the stack heats up. As the focus of this project is on electrical control rather than thermal management, this is considered an acceptable limitation.

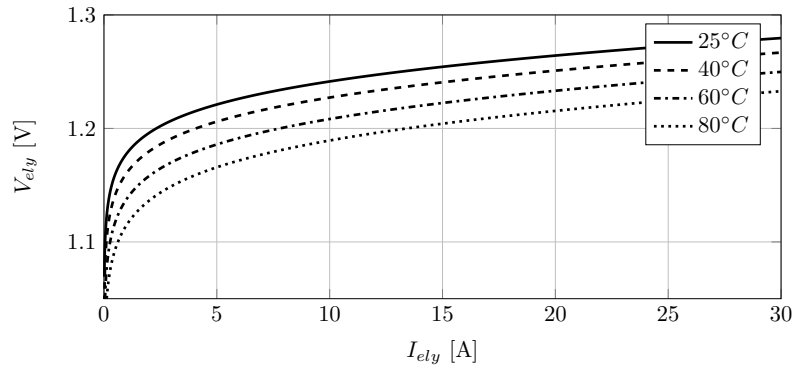


Figure 2.3: Polarisation curve of Proton Exchange Membrane Electrolyser at different temperatures

2.1.6 Circuit Model

As the electrolyser is a load, it needs to be modelled as such in order to include it in a larger electrical system such that dynamic relationships between system components are accounted for. There are two general methods for creating a circuit equivalent of an electrolyser, a dynamic model, and a static model. The static model only aims to replicate the polarisation curve and steady state operation and is quite viable for longer time frames such as hours or days. In these cases there is little need for the short term electrical dynamics of the system as they would be masked by the large time-steps required. However, in this project the goal was to investigate control methods, and thus, a dynamic model is required.

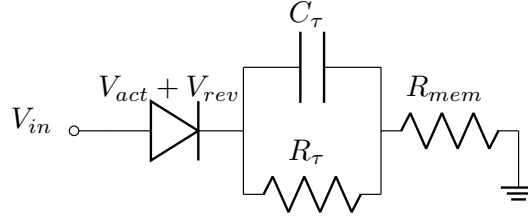


Figure 2.4: Equivalent electrical circuit representation of PEM electrolyser featuring accurate steady state and dynamics. A spice type diode is vital for representation as it features logarithmic like increase in current draw as voltage increases in contrast to piece wise diodes which are either open, or closed with a constant voltage drop

At first, a static baseline model was made to make sure the IV response was accurate. For this, a spice diode in series with a resistor was chosen. The diode models the activation and reversible voltage by only allowing current to pass past a certain threshold and by having a curved response to current beyond said point, in comparison to a step-wise diode which has two modes, open and closed circuit with a constant voltage drop. As the membrane resistance is already ohmic, it was modelled as a resistor with a fixed impedance. Both of these components were then tuned against a reference static physical model at with a stack count of $N = 50$ cells.

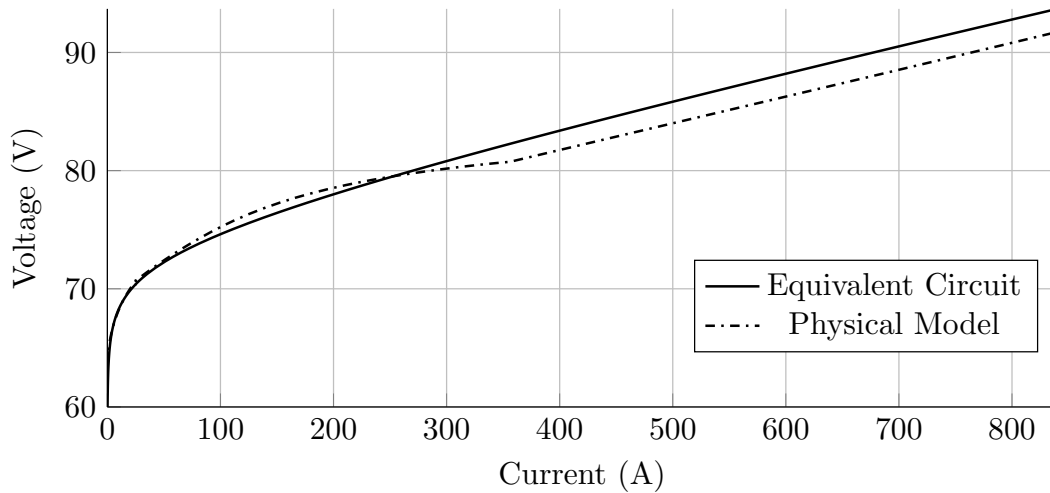


Figure 2.5: steady state IV response of physical model and circuit equivalent simulated using Simulink

Tuning the values through trial and error gave the values found in Table 2.1. The high emission coefficient of the diode leaves a slow slope which very closely resembles that of the physical model as seen in Figure 2.5. The physical model exhibits an anomalous discontinuity in slope around 350A which could not be explained

by any physical phenomena or by any of the overpotential terms in the electrochemical model and is therefore attributed to a modelling artefact of the physical model [21].

Name	Value
Emission Coefficient	76
$R_{mem,stat}$	0.02Ω

Table 2.1: Steady state PEM circuit equivalent values

The emission coefficient of 76 is far outside the range of physically possible diodes and is to be understood as an empirical fitted parameter rather than a physical property. The diode used here is purely for modelling purposes to achieve the curved I/V characteristic which approximates the combined activation and reversible voltages of the polarisation curve.

After the static model had been defined, the dynamics could be built on top of it. The short-term dynamics of the electrolyser are one or two time constants depending on whether both electrodes are modelled as driving the dynamics. For this project just one time constant was chosen by including a large capacitor (C_τ) in the circuit. The resistive element ($R_{mem,stat}$) was split into two, one resistor (R_{mem}) in series with the capacitor and another resistor (R_τ) in parallel (Fig. 2.4). This configuration also allows for tuning of the dynamic response without altering the steady state response by adjusting the ratio of impedance between R_{mem} and R_τ to alter the charge time of the capacitor. As long as the sum of the resistors are kept at 0.02Ω , the steady state response remains the same. The capacitor was tuned to $37F$. [18]

Name	Value
Emission Coefficient	76
Diode Impedance	$10^{-6}\Omega$
R_{mem}	0.01Ω
R_τ	0.01Ω
C_τ	$37F$

Table 2.2: Dynamic PEM circuit equivalent values

2.2 Converters

2.2.1 Rectifier

In order to power the electrolyser, DC (Direct Current) power is needed, meaning the current travels one way, compared to AC (Alternating Current) where the current travels back and forth following a sinusoidal shape. Rotating generators such as turbine based nuclear reactors, coal reactors and wind turbines all generate an alternating current, which is the standard used for the entire grid. To convert alternating current to a direct current, a rectifier is needed. Rectifiers alter the nature of an AC signal such that it becomes a relatively clean DC signal by utilising diodes to move current one way only. For single phased AC, a full diode bridge is most commonly utilised.

Full-bridge rectifier

The full diode bridge is designed such that any positive current from the AC is passed through D_2 and D_3 , while any negative currents are passed through D_4 and D_1 . This creates a pathway for both the positive

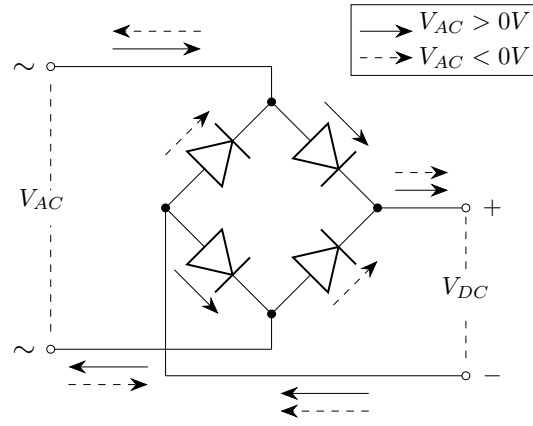


Figure 2.6: Single-phase full bridge rectifier

and negative cycle of the supply in which they are channel in such a way that they both move from positive to negative at the output of V_{DC} despite moving in opposite directions at V_{AC} (Fig. 2.6).

Due to diode forward voltage, diode rectifiers feature losses from near constant voltage drops. For many hobby diodes, this voltage drop is usually in the range of $0.3V - 0.7V$, and since each path passes through two diodes, the overall voltage drop of the circuit becomes double this ($0.6 - 1.4V$). Often in cases where the losses from the diodes are a large issue, switching rectifiers are favoured, where instead of passive diodes, controlled MOSFETS are used, using sensors and logic to close S_1 and S_4 when the current is positive, and S_2, S_3 when the current is negative. This allows for much lower losses, but does require additional circuitry and logic, which drives up cost.

Three-phase full-bridge rectifier

The grid features three lines of $50Hz$ each with their phase separated 120° apart. Most household appliances only connect to one phase, however, industrial applications usually use all three phases, and thus, need a way to rectify the current if the device requires a direct current like is the case with electrolyzers.

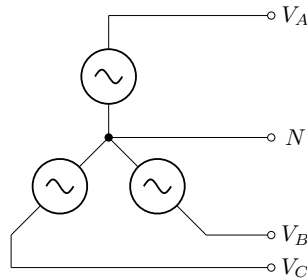


Figure 2.7: Three-phase power source in Wye-configuration. Allows for phase-to-neutral connections to get $230V_{rms}$ or phase-to-phase connections to achieve $400V_{rms}$ depending on application need and stability concerns

In this project the source was chosen as a simple three-phase voltage source (fig. 2.7). This source can be altered to represent that of a grid connection, a wind turbine or a generator depending on the voltage and current output, as well as source impedance. For three phases, it's necessary to use a suitable three-phase rectifier. For simplicities sake, the rectifier in question the passive three-phase full bridge diode rectifier.

The three-phase rectifier consists of three pairs of two diodes, each pair is connected to a phase, and the common positive and negative terminal of the output. The overall structure is almost identical to that of the full bridge rectifier, with only one additional pair of diodes added. However, the output from the three-phase rectifier has much lower ripple than its full-bridge counterpart. A smoothing capacitor is added

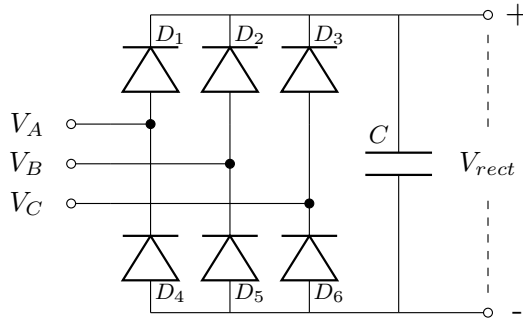


Figure 2.8: Conventional three-phase diode bridge rectifier configuration with a diode pair for each path and a smoothing capacitor on the output to reduce ripple.

in parallel to the output to supply any sudden increases in current draw and stabilize the output further.

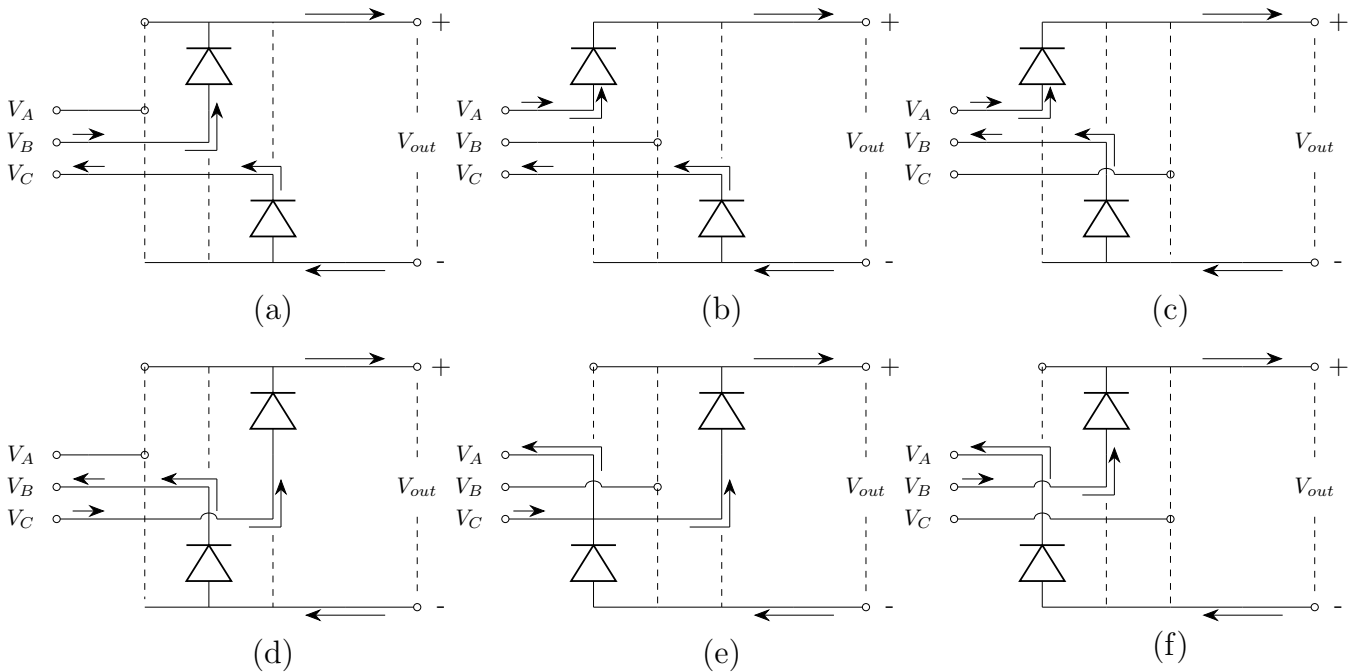


Figure 2.9: Conduction stages (a)–(f) of the three-phase full-bridge rectifier, showing the six intervals of the AC cycle. At any given moment, the phase with the highest instantaneous potential overcomes both the diode forward voltage and V_{out} , conducting on the positive rail, while the phase with the lowest instantaneous potential conducts on the negative rail. The remaining phase does not have sufficient potential to overcome its diode and does not contribute. As the phases alternate in dominance across the cycle, the conducting pair rotates through the sequence (a) $V_B \rightarrow V_C$, (b) $V_A \rightarrow V_C$, (c) $V_A \rightarrow V_B$, (d) $V_C \rightarrow V_B$, (e) $V_C \rightarrow V_A$, (f) $V_B \rightarrow V_A$.

The three-phased full bridge rectifier operates as seen in figure 2.9 in six stages. At any time, one phase is positive and is supplying current to the positive line, one phase is negative and is drawing current from the negative line, and one does not have a high enough voltage such that it can bypass its diode and does not contribute to the output. As the phases oscillate as seen in figure 2.10, their role changes. The resulting output power is more stable than that of the single phase rectifier simply by the nature of having more phases.

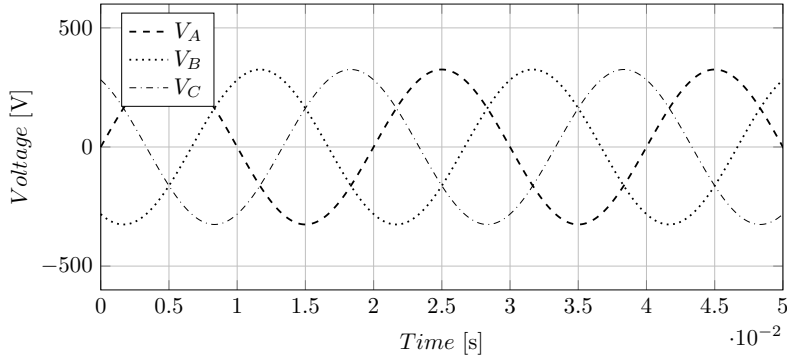


Figure 2.10: Grid Source

Figure 2.11 shows the resulting output of the three-phase rectifier. Due to the rectifier inverting the negative parts of the phases and outputting the instantaneous phase-to-phase voltage of the two currently conducting phases (Fig. 2.9), the effective frequency of the output is essentially six times that of the fundamental ($50Hz$). It can also be seen from the graph that the actual output lies higher than that of the peak voltage of the three-phase system. This is again due to the rectifier passing the voltage as phase-to-phase, meaning the reference point at any time is not the neutral voltage, but the current lowest phase. This yields a increase in the output voltage of $\sqrt{3} = 1.732$ times that of the peak of any phase relative to neutral. This results in the three phases each supplying $230V_{rms}$, which has an instantaneous peak of $325V_{pp}$, and outputting roughly $540V_{dc} - 566V_{dc}$ when rectified as a three phase system depending on the smoothing capacitor.

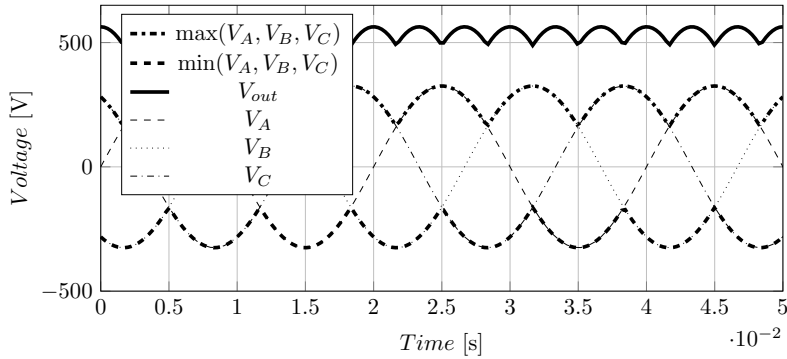


Figure 2.11: Operation of Three-Phase rectifier prior to capacitor smoothing

This voltage level is quite a lot higher than the rating of the electrolyser in question which will be run at $100kW$ around $100V$. It's clear that a way to step down the voltage is required in order to stay within operating limits. For this, a buck converter makes the simplest addition to the circuit.

2.2.2 Buck Converter

Often, the operator of an electrolyser wants to be able to control the amount of current, voltage or power draw of the device. Additionally, electrolyzers are low voltage - high current devices, which does not match with the grid, favouring high voltage for long distance transportation of power. Both of these issues are tackled by the controlled DC-DC converter. A converter is a device that takes either an AC or DC voltage, and outputs an AC or DC voltage that's higher, lower or equal to the input. As the grid operates at high-

voltage and electrolyzers operate at low voltage, a DC-DC converter designed for stepping down the voltage is required, such device is known as a "buck" converter.

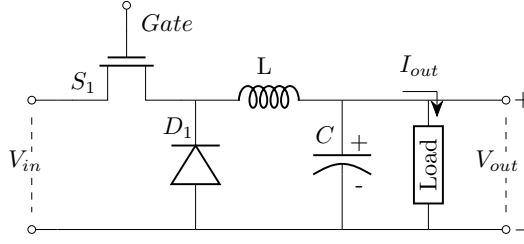


Figure 2.12: Conventional buck converter topology with switch, diode, inductor and output capacitor and load

The buck converter is a switching DC-DC converter, meaning it performs the conversion by opening and closing the connection to the power source. It operates by storing electrical energy as a magnetic field in the inductor when the switch is closed, and draining the inductor when the switch is open. This keeps current flowing through the load while the switch is either open or closed. A diode is placed in parallel with the load behind the inductor and acts like a valve, creating a complete circuit when the switch is open, without letting current pass from a closed switch to ground, causing a short. To smooth the output voltage further, a capacitor is placed in parallel with the load, storing any power not instantly utilised by the load and returning it as the inductor discharges.

The device is controlled using a PWM signal, which is a high frequency binary signal with variable duty cycle, for duty cycles higher than 50%, the switch stays closed for longer than it stays open, allowing more time for the inductor to charge, and thus, increasing the voltage output. This relationship between input voltage, duty cycle and output voltage is largely linear, meaning an input of 100V at a duty cycle of 50% results in an output of $\sim 50V$, leading to the calculation for duty cycle (Eq. 2.13)

$$d = \frac{V_{out}}{V_{in}} \quad (2.13)$$

Simply setting a duty cycle however does not guarantee the correct output voltage, current or power, as such, feed-back control is necessary in order to confidently reach the desired output. Which is why converters like these need to be either tuned to the correct output, or actively controlled, as will be the case in this project.

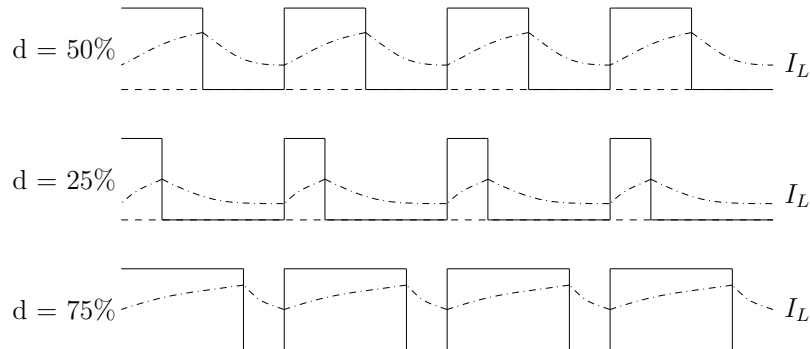


Figure 2.13: Charging and discharging cycles of buck converter at different switching duty-cycles, dashed-dotted line shows instantaneous current output of the buck I_L while full lines show the instantaneous switch status (open or closed)

Figure 2.13 shows exaggerated operation of buck converter with 50% duty cycle resulting in the inductor current staying mostly in the middle, while 25% brings it much lower and 75% bringing it much higher.

Since the inductor discharges quicker at high values and charges faster at low values, it can be seen that the slope at 75% when discharging is much steeper than that of the slope when charging, same is seen for 25%. This is what enables switching to even work, as a system with identical slopes would eventually settle at 100% or 0% output at duty cycles other than 50%. The current flow of the charging and discharging events can be seen in figure 2.14.

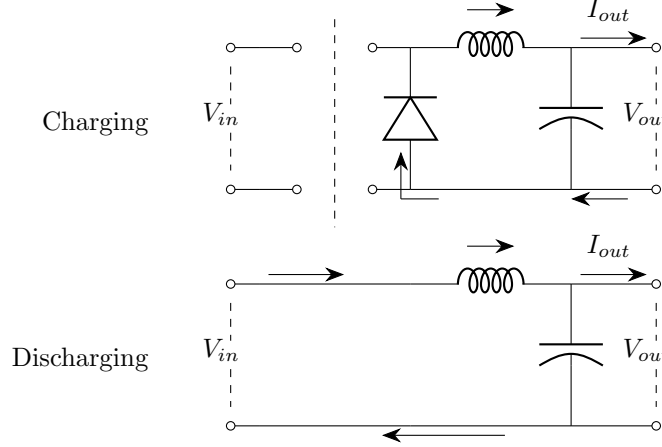


Figure 2.14: Buck converter charging and discharging equivalent circuit topology showing current flow of open and closed switch state

The standard buck converter uses a single switch, which results in large charge and discharge cycles, causing large ripple currents which can degrade the electrolyser over time. The single switch also supplies all the current for the entire electrolyser stack, and as the electrolyser is essentially a low impedance load, the current passing through the one switch is often very high and would over time stress the switch to the point of failure. To solve these issues, this project makes use of a three-phase interleaved buck converter instead. The interleaved buck utilises multiple separately complete functioning buck converters, each running at the same duty cycle. The only difference between each converter is the phase which the of the gate signal. For a three-phase converter, each gate signal is offset from the other by 120° . The current output of the full converter is the same as with a single buck, which thus means that each switch is only subject to $\frac{1}{N}i_{out}$ where N is the amount of bucks in the system. This helps with mitigating degradation of the switches by not letting a single switch take the entire load by itself, and also adds redundancy, as any of the three switches can break in such a way that it becomes an open circuit, without the entire system breaking down. In this case, the two buck converters left will now be subject to $\frac{1}{N-1}i_{out}$.

The three-phase interleaved buck also achieves lower output current ripple due to the switching being offset between phases. Each buck stage has its own charge and discharge cycle, but as with the three-phase rectifier (Sec. 2.2.1), the output current is the sum of all three phase currents, resulting in lower variation and a ripple frequency three times that of a single phase. The effective output inductance of the interleaved converter is also reduced to a third of the per-phase inductance, since the three inductors appear in parallel from the perspective of the output. A lower effective inductance allows the inductor current to slew more rapidly in response to changes in input voltage, giving the converter a faster transient response under variable source conditions such as those produced by renewable energy sources.

There are some drawbacks to using this system however, as the additional components, switches, PWM generators and logic result in an overall higher prices, but for this project, the more stable output and the less stain on the switches themselves was deemed to largely outweigh the downsides of adding complexity and price.

The interleaved buck converter requires no additional control logic aside from phase shifted PWM generators

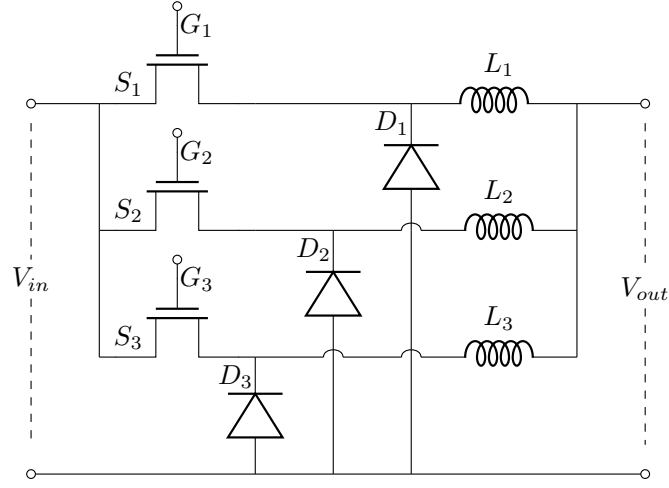


Figure 2.15: General topology of three-phase interleaved buck converter featuring three individual buck circuits with switches, inductors and diodes

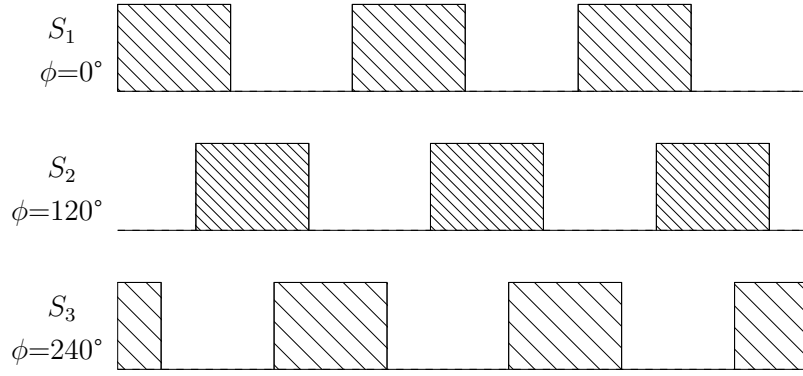


Figure 2.16: Gate signals of three-phase interleaved buck converter at 50% duty cycle

Figure 2.16 shows the control signal of each switch at a 50% duty cycle. Each phase is seen as having their own phase-shifted version of the same gate signal.

Deciding on the values of the buck was done in the same manner as with a regular buck converter, where operating point values are decided alongside voltage and current ripple maximum to numerically calculate the components.

Variable	Symbol	Value
Phases	N	3
Input Voltage (DC)	V_{in}	566V
Output Voltage (DC)	V_{out}	100V
Switching frequency	f_s	1kHz
Rated Power (Output)	P	100kW
Ripple current (Per Phase)	ΔI_L	10%
Ripple voltage (Output)	ΔV_{out}	1%

Table 2.3: Buck Converter specifications

The equations differ slightly for interleaved buck converters since the average current seen by the inductors and the capacitor are different, and thus, so is the ripple. First the values of the inductors are calculated

Variable	Value
$L_{1,2,3}$	$2.47mH$
C_b	$870\mu F$

Table 2.4: Buck converter component values

with the assumption that the average current across them is a third of the overall current of the system.

$$d = \frac{V_{out}}{V_{in}} = 17.66\% \quad (2.14)$$

$$I_{out} = \frac{P}{V_{out}} = 1000A \quad (2.15)$$

$$I_L = \frac{I_{out}}{3} = 333A \quad (2.16)$$

$$\Delta I_L = 333A \cdot 10\% = 33.3A \quad (2.17)$$

$$L = \frac{d \cdot (V_{in} - V_{out})}{\Delta I_L \cdot f_s} = 2.47mH \quad (2.18)$$

The inductor is thus calculated to be around $2.47mH$. The duty cycle has to remain quite low in order to step down the voltage of the system to the $100V$ target. This value turns out to be less than ideal for ripple cancellation due to the interleaved nature, since the cancellation coefficient K only attenuates the ripple by a factor 0.6.

$$K = 1/N \cdot \left| \frac{\sin(N\pi d)}{\sin(\pi d)} \right| = 0.627 \quad (2.19)$$

Using the cancellation coefficient the effective ripple at the output capacitor is found. The capacitor can be calculated in the same way that a single phase buck usually is with the change that the ripple is effectively less, and due to the three phases, is subjected to an effective frequency of $3 \cdot f_s$

$$\Delta I_{out} = K \cdot \Delta I_L = 0.627 \cdot 33.3A = 20.87A \quad (2.20)$$

$$C_b = \frac{\Delta I_{out}}{8 \cdot N f_s \cdot \Delta V_{out}} = 870\mu F \quad (2.21)$$

2.3 Mathematical Model

In order to control the system properly, a mathematical model of the system dynamics was developed. This section derives the model and converts it to a state-space representation and transfer function suitable for defining the MPC controller. Each subsystem is defined separately and connected through common variables such as the electrolyser voltage and the inductor current of the buck converter.

2.3.1 Three-phase full-bridge Rectifier

The first element to define is the rectifier. As described in Section 2.2.1, the chosen topology is the three-phase full-bridge diode rectifier, which features six diodes arranged such that at any point in time one phase is conducting on the positive line and one on the negative. The mathematical model used here is the average model, in which the 300 Hz output ripple produced by the six-pulse rectification is neglected.

This simplification removes unnecessary poles and zeros from the plant model. The trade-off is that ripple dynamics and source impedance effects are not captured by the model, the output of the rectifier will be considered a constant voltage source upstream. The average output voltage is given by Equation 2.22, where $V_{in_{rms}}$ is the phase-to-phase RMS source voltage, which for a grid-connected system is approximately $400 V_{rms}$, yielding a DC output of 540 V.

$$V_{out_{avg}} = V_{dc} \approx \frac{3\sqrt{2}}{\pi} V_{in_{rms}} \approx 1.35 \cdot 400V = 540V \quad (2.22)$$

Since the rectifier has no dynamics in this model, the DC bus voltage is treated as a constant. Though the calculation gave 540V, a smoothing capacitor would allow the system to run closer to 566 V, which will thus be the voltage level of the source for the remainder of the system. In practice, source impedance may cause small deviations from this value, but the simplification is stays within the assumptions previously established.

2.3.2 Buck converter

The buck is the first part of the system which introduces dynamics and non-linearities by featuring both an inductor and a smoothing capacitor. The buck will also be modelled as an average, but because it contains dynamics, those need to be factored into the equation. At first, the steady voltage across the inductor at the two states of the buck (open, closed) will be determined.

$$V_{L_{closed}} = V_{dc} - v_o \quad (2.23)$$

$$V_{L_{open}} = -v_o \quad (2.24)$$

When the switch is closed and current is passing through the buck, the inductor acts like a threshold with the DC voltage calculated from the rectifier on one side, and the electrolyser voltage on the other. When the switch is open, the side that connected to V_{dc} is now connected directly to ground, and thus, the voltage across the inductor is simply $-v_o$. All diodes in this model are assumed to be lossless closed circuit when conducting.

Since the gate signal switches between these two states at a ratio equal to the duty cycle, the average model can be determined by saying that the output voltage must be a linear combination of the two states at a ratio equal to the duty cycle (EQ. 2.25).

$$\begin{aligned} V_L &= d \cdot (V_{dc} - v_o) + (1 - d)(-v_o) \\ &= d \cdot V_{dc} - v_o \end{aligned} \quad (2.25)$$

With the average inductor voltage defined, the inductor current can be derived simply as in equation 2.26.

$$\frac{di_L}{dt} = \dot{i}_L = \frac{1}{L}(d \cdot V_{dc} - v_o) \quad (2.26)$$

Here the derivative is marked as $\dot{i}_L$, since it is considered a state of the state-space representation of the circuit. The inductor current splits at the output and runs through both the electrolyser and the smoothing capacitor, this dynamic can be defined as equation 2.28, which is the formula for capacitor voltage.

$$i_L = i_{C_b} + i_{ely} \quad (2.27)$$

$$i_{C_b} = C_b \frac{dv_o}{dt} \quad (2.28)$$

Substituting I_C and solving for the derivative of the output voltage from equation 2.28 yields the second state equation (EQ. 2.29).

$$\begin{aligned}
i_L &= C_b \frac{dv_o}{dt} + i_{ely} \\
C_b \frac{dv_o}{dt} &= i_L - i_{ely} \\
\frac{dv_o}{dt} &= \dot{v}_o = \frac{1}{C_b} (i_L - i_{ely})
\end{aligned} \tag{2.29}$$

The last part of the mathematical model is the PEM electrolyser.

2.3.3 Electrolyser

Despite the circuit equivalent derived in section 2.1.6 utilising a diode to represent the activation and reversible overpotentials. The mathematical model will use the actual calculations for these values while still having the split resistor setup such that the capacitor can be included in the calculations. In place of the diode a voltage source (Eq. 2.30) is placed facing backwards such as to act as a current controlled voltage drop.

$$V_n(i_{ely}) = V_{rev} + V_{act}(i_{ely}) \tag{2.30}$$

As $V_n(i_{ely})$ is current dependent the current running through the electrolyser i_{ely} is to be defined. First, the current running through the capacitor of the dynamic part of the electrolyser is defined as the derivative of the voltage drop across both R_τ and C_τ in parallel multiplied by the capacitance of C_τ (Eq. 2.33).

$$i_c = i_{ely} - i_r \tag{2.31}$$

$$i_r = \frac{v_\tau}{R_\tau} \tag{2.32}$$

$$i_c = C_\tau \cdot \frac{dv_\tau}{dt} \tag{2.33}$$

With the capacitor current defined it's possible once again to solve for the derivative of the voltage across said capacitor to get Eq. 2.34. For which i_c and i_r can be substituted by 2.31 and 2.32 to get the third state of the system (Eq. 2.35).

$$\frac{dv_\tau}{dt} = \dot{v}_\tau = \frac{1}{C_\tau} \cdot i_c \tag{2.34}$$

$$\dot{v}_\tau = \frac{1}{C_\tau} \cdot \left(i_{ely} - \frac{v_\tau}{R_\tau} \right) \tag{2.35}$$

$$\begin{bmatrix} \dot{i}_L \\ \dot{v}_o \\ \dot{v}_\tau \end{bmatrix} = \begin{bmatrix} \frac{1}{L} (d V_{rec} - v_o) \\ \frac{1}{C_b} (i_L - i_{ely}) \\ \frac{1}{C_\tau} \left(i_{ely} - \frac{v_\tau}{R_\tau} \right) \end{bmatrix} \tag{2.36}$$

With the states defined the last missing part is to define i_{ely} and include $V_n(i_{ely})$ as well as R_{mem} in the state-space model. This is done by first defining voltage drop across U_{mem} as seen in equation 2.37. From there, the voltage at the output of the buck converter v_o can be defined as the combination of the voltage drop of the three sections of the electrolyser circuit equivalent (Eq. 2.38) and inverted as to derive a definition for i_{ely} which can be substituted.

$$v_{mem} = R_{mem} \cdot i_{ely} \tag{2.37}$$

$$\begin{aligned}
v_o &= V_n(i_{ely}) + v_\tau + v_{mem} \\
i_{ely} &= V_n^{-1}(v_o - v_\tau - v_{mem}) \\
i_{ely} &= g(v_o - v_\tau - v_{mem})
\end{aligned} \tag{2.38}$$

Due to the complexity of $V_n(i_{ely})$ it's infeasible to algebraically define $V_n^{-1}(v)$, however, it will later be numerically computed once an operating point is defined. With the three state variables now defined (EQ.2.36), It's time to linearise and compute the state space representation.

Since V_n is highly non-linear it can be linearised near the operating point. Essentially computing the tangent at the operating point, which will be relatively accurate within reason for small perturbations in the system. The method used to derive this linearised equation is the Taylor series (EQ. 2.39) Which defines an approximation $T(x)$ of a function ($f(x)$) as the sum of derivatives of said function divided by the factorial of same size as the derivative, multiplied by the difference between a designed operating point z_0 and the actual current point of operation z .

$$T(z) = f(z_0) + \frac{f'(z_0)}{1!}(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \dots + \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n \tag{2.39}$$

Since the system is being linearised, there is no need for any more than the second term, as linear functions only feature slope and thus the equation becomes

$$T_{lin}(z) = \frac{f'(z_0)}{1!}(z - z_0) = \frac{g'(z_0)}{1!}(z - z_0) \tag{2.40}$$

The operating point z_0 can be defined as $z_0 = (v_{o0} - v_{\tau0})$, the current point as $z = (v_o - v_\tau)$ and the difference between the two $\tilde{z} = z - z_0$ such that the linearised approximation becomes

$$\tilde{i}_{ely} \approx g'(z_0)\tilde{z} \tag{2.41}$$

To make it easier to read the state space matrices, $g'(z_0)$ will be represented as simply G . And since the system is now linearised for small perturbations around the an operating point, the other variables will be represented in the same manner as $\tilde{z}$

$$\tilde{i}_L = i_L - i_{L0}, \quad \tilde{v}_o = v_o - v_{o0}, \quad \tilde{v}_\tau = v_\tau - v_{\tau0} \tag{2.42}$$

$$\begin{bmatrix} \dot{\tilde{i}}_L \\ \dot{\tilde{v}}_o \\ \dot{\tilde{v}}_\tau \end{bmatrix} = \begin{bmatrix} \tilde{d} \frac{V_{rec}}{L} - \frac{\tilde{v}_o}{L} \\ \frac{\tilde{i}_L}{C_b} - \frac{G}{C_b}(\tilde{v}_o - \tilde{v}_\tau) \\ \frac{G}{C_\tau}(\tilde{v}_o - \tilde{v}_\tau) - \frac{\tilde{v}_\tau}{C_\tau \cdot R_\tau} \end{bmatrix} \tag{2.43}$$

With the voltage source symbolically linearised, the state space matrices can be derived, the vector $\tilde{x}$ contains the state variables, the variable $\tilde{u}$ represents the control input (duty cycle), $\mathbf{A}$ contains the interactions between said variables, $\mathbf{B}$ maps the control input to the states, and $\mathbf{C}$ is the output matrix, $\tilde{y}$ maps the state vector to the system output $\tilde{y}$

$$\dot{\tilde{x}} = \mathbf{A}\tilde{x} + \mathbf{B}\tilde{u} \tag{2.44}$$

$$\tilde{y} = \mathbf{C}\tilde{x} \tag{2.45}$$

$$\tilde{x} = \begin{bmatrix} \tilde{i}_L \\ \tilde{v}_o \\ \tilde{v}_\tau \end{bmatrix}, \quad \tilde{u} = \tilde{d} = d - d_0 \tag{2.46}$$

Calculating each matrix the following matrices are derived as the state space representation of the entire circuit:

$$\mathbf{A} = \begin{bmatrix} 0 & -\frac{1}{L} & 0 \\ \frac{1}{C} & -\frac{G}{C} & \frac{G}{C} \\ 0 & \frac{G}{C_\tau} & -\left(\frac{G}{C_\tau} + \frac{1}{R_\tau C_\tau}\right) \end{bmatrix}, \mathbf{B} = \begin{bmatrix} \frac{V_{rec}}{L} \\ 0 \\ 0 \end{bmatrix} \quad (2.47)$$

$$\mathbf{C} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}, \quad (2.48)$$

2.3.4 Defining the operating point

The operating point is the point at which for small perturbations around it, the approximation is the closest to it's original function, it is where the tangent will be placed, and as such, it needs to be picked such that for the most part, the electrolyser will be operating around said point. Since the duty cycle for the switches in the buck is what is controlled, it can be tuned in a feed forward configuration such that it at steady state yields the target of $100kW$ on the electrolyser. Which also means that the mathematical model itself needs to be recalculated such that it has no dynamics and is in steady state. This is done by setting the derivatives of the state variables to zero and solving the three equations as such (2.49, 2.50, 2.51).

$$0 = \frac{1}{L}(d_0 V_{rec} - v_{o0}) \quad (2.49)$$

$$v_{o0} = d_0 \cdot V_{rec}$$

$$0 = \frac{1}{C_b}(i_{L0} - i_{ely0}) \quad (2.50)$$

$$i_{L0} = i_{ely0}$$

$$0 = \frac{1}{C_\tau} \left(i_{ely0} - \frac{v_{\tau_0}}{R_\tau} \right) \quad (2.51)$$

$$v_{\tau_0} = R_\tau \cdot i_{ely0}$$

Since the calculations for the buck components computed a duty cycle of $d = 17.66\%$ to achieve the $100V$ (Eq. 2.2.2), the same value will be used here.

$$v_{o0} = 17.66\% \cdot 566V \approx 100V \quad (2.52)$$

The average model used to calculate v_{o0} has a lot it doesn't take into account, for example the fact that a small load may drain the inductors faster and bring the overall voltage level down, however, it is good enough that it can be used here. Equation 2.50 shows that at steady state, the inductor is a short, and thus the current across the inductor is identical to that of the electrolyser branch, thus can thus be ignored and the last state v_{τ_0} can be calculated.

$$v_{\tau_0} = R_\tau \cdot i_{ely0} = 0.01\Omega \cdot i_{ely0} \quad (2.53)$$

Here, the value of the resistor R_τ derived from table 2.2 is used. The split resistor equivalent is used as the voltage drop v_{τ_0} is only present across R_τ and not also R_{mem} .

The activation and reversible voltages in $V_n(i)$ can be derived as well using the equations 2.1.2 & 2.1.1, constants R and F are already known as the gas law and Faraday law respectively, the temperature has been defined to be $80^\circ C$ at steady state, and from the reference model, current density i_0 and cells area A can be found to be $1e-4$ and $5.6cm^2$ respectively. Both values are multiplied by the cell count such that they represent the entire stack and not just a single cell.

$$\begin{aligned}
U_{act_0} &= \frac{R \cdot T}{\alpha Z \cdot F} \cdot \ln \left(\frac{i}{i_0} \right) \cdot N \\
&= \frac{8.314 \cdot 353.15}{0.5 \cdot 2 \cdot 9.648 \cdot 10^4} \cdot \ln \left(\frac{i_{ely_0}}{1 \cdot 10^{-4}} \right) \cdot 50 \\
&\approx 1.522 \cdot \ln \left(\frac{i_{ely_0}}{0.0001} \right)
\end{aligned} \tag{2.54}$$

From the same assumptions of the system, the reversible voltage can be derived simply by substituting the temperature in equation 2.1.1.

$$\begin{aligned}
U_{rev} &= (1.5184 - 1.5421 \cdot 10^{-3} \cdot T \\
&\quad + 9.523 \cdot 10^{-5} \cdot T \cdot \ln(T) \\
&\quad + 9.84 \cdot 10^{-8} \cdot T^2) \cdot N \\
&= 1.2291 \cdot N \\
&= 59.169V
\end{aligned} \tag{2.55}$$

Since the entire electrolyser stack was previously defined in equation 2.38 as the combination of the voltage drops for each of the components of the circuit, the current through the entire system can now be calculated by substituting the values and expressions calculated in this chapter and solving for i_{ely} .

$$\begin{aligned}
v_{o_0} = 100V &= U_n(i_{ely_0}) + v_{\tau_0} + U_{mem_0} \\
&= U_{rev} + U_{act_0} + v_{\tau_0} + R_{mem} \cdot i_{ely_0} \\
&= 59.169 + 1.522 \cdot \ln \left(\frac{i_{ely_0}}{0.0001} \right) + 0.01 \cdot i_{ely_0} + 0.01 \cdot i_{ely_0} \\
&= 59.169 + 1.522 \cdot \ln \left(\frac{i_{ely_0}}{0.0001} \right) + 0.02 \cdot i_{ely_0}
\end{aligned} \tag{2.56}$$

$$i_{ely_0} \approx 1000 \text{ A} \tag{2.57}$$

Which confirms that for this model, a duty cycle of 17.66% yields a power output of 100kW From here G can be calculated.

G was defined in Eq. 2.38 and 2.39 as the derivative of the inverse of the polarisation curve. To make matters easier, the derivative can be calculated first such that i_{ely_0} can be substituted for the explicit value just calculated, and the numerical answer then inverted as detailed in equation 2.58

$$G = g'(v_{o_0} - v_{\tau_0}) = \frac{d}{dv} V_n^{-1}(v_{o_0} - v_{\tau_0}) \rightarrow \frac{1}{V'_n(i_{ely_0})} \tag{2.58}$$

The derivative cancels out all the static offset of the reversible voltage and the non-linear logarithmic term such that what is left is purely linear, this term can then be inverted to arrive at a numerical value for G which can be substituted in the state space to finally arrive at a transfer function of the system as a whole.

$$\begin{aligned}
V'_n(i_{ely_0}) &= \frac{d}{di_{ely_0}} \left(59.169 + 1.522 \cdot \ln \left(\frac{i_{ely_0}}{0.0001} \right) + 0.02 \cdot i_{ely_0} \right) \\
&= \frac{1.521}{i_{el_0}} + 0.02
\end{aligned} \tag{2.59}$$

$$\frac{1}{V'_n(i_{el_0})} = \frac{1}{0.021521\Omega} = 46.4A/V \tag{2.60}$$

2.3.5 Calculating the transfer function

With all the values required for calculating the transfer function derived. It's possible to substitute each in the state space matrices and use python to compute the transfer function of the system, which can be used to control the system using the MPC controller. Notice that the inductor value L in table 2.5 is $823\mu H$ and not $2.47mH$ as defined in section 2.2.2. This is due to the fact that for an interleaved buck, the effective inductance of the system is equal to L/N with N being the about of phases in said converter. As such, the effective inductance of the buck in this system is $0.823\mu H$.

Table 2.5: Transfer Function Values

Symbol	Value
v_{o0}	100V
G	46.4A/V
L	823 μH
C_b	870 μF
C_τ	37F
R_τ	10m Ω

$$A = \begin{bmatrix} 0 & -\frac{1}{823 \cdot 10^{-6}} & 0 \\ \frac{1}{870 \cdot 10^{-6}} & -\frac{46.4}{870 \cdot 10^{-6}} & \frac{46.4}{870 \cdot 10^{-6}} \\ 0 & \frac{46.4}{37} & -\left(\frac{46.4}{37} + \frac{1}{0.01 \cdot 37}\right) \end{bmatrix} \quad (2.61)$$

$$B = \begin{bmatrix} \frac{566}{823 \cdot 10^{-6}} \\ 0 \\ 0 \end{bmatrix} \quad (2.62)$$

To get the transfer function the `control` library from python was used, which allows defining state space systems using `numpy` matrices and converting said systems into transfer functions using the method `ctrl.ss2tf()`.

$$\frac{4.366e-11s^2 + 7.905e+8s + 3.129e+9}{s^3 + 5.341e+04s^2 + 1.541e+06s + 5.529e+06} \quad (2.63)$$

This resulted in the following transfer function (Eq. 2.63). To check the accuracy of the system a step response plot was computed, step responses are usually from 0 – 1, however since the system was calculated from the assumption that the duty cycle was $d = 0.18$, the step response was made under same assumptions, and since the transfer function is linear, it was possible to simply multiply the y axis of the plot by the duty cycle to get the proper sized plot.

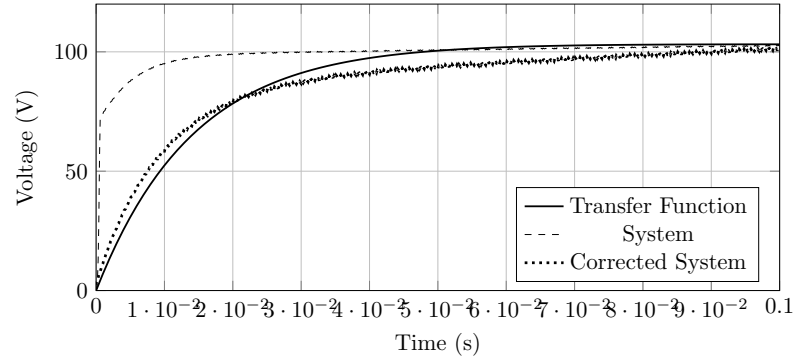


Figure 2.17: Comparison between uncontrolled dynamics of derived transfer function and circuit equivalent system when subjected to step of 100V. Circuit equivalent is both shown moving from 0 to 100, and 100 to 200 (corrected) such as to show the actual dynamics without the interference of the diode emission coefficient which prevents current from passing before approximately 75V

From Figure 2.17 it can be seen that the transfer function and the actual system doesn't seem to line out quite right, however, if corrected for the non-linearity of the reversible and activation voltages being omitted from the transfer function — which is the reason why the system rises to 75V quickly in the beginning — by stepping from 100 to 200, and fitting it to 0 – 100, it can be seen that the dynamics of the two are almost identical for $t = 0.1s$. Only the magnitude of the system is affected by the heavy non-linearity, of the electrolyser, the dynamics for the most part are close to linear.

2.4 Simulation Setup

The system was build in Simulink using the simscape electrical package. For simple components and sensors said package was used exclusively, however for complex parts like the fault subsystem, rectifier, buck converter and the electrolyser, built-in Simulink blocks were omitted in favour of component level representation. This was done partly to have more control over the individual parts of the simulation, and due to Simulink blocks often having large levels of abstraction and only working under very specific conditions like will be discussed in chapter 4.

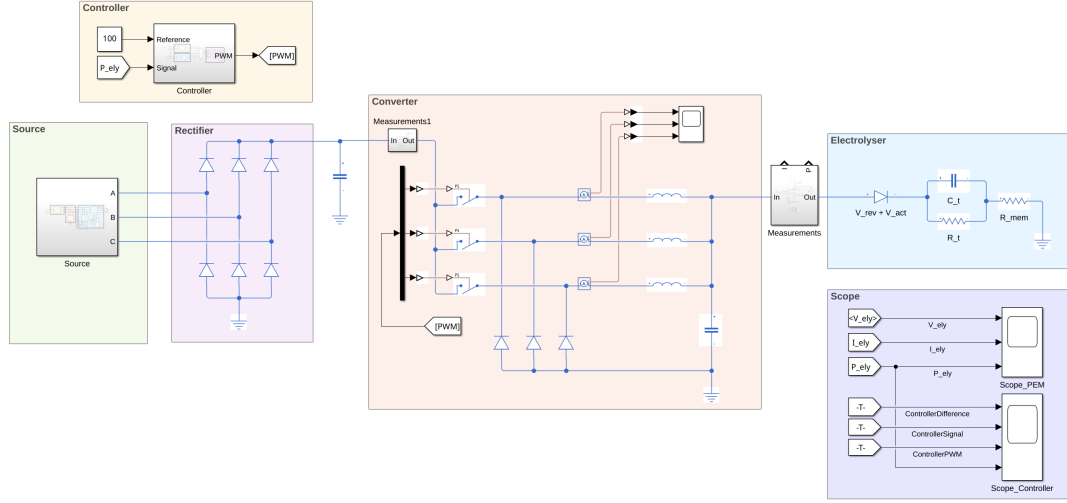


Figure 2.18: Simulation setup from the top level, features rectifier circuit, interleaved buck converter with controller and circuit equivalent of PEM hydrogen electrolyser as well as temperature estimation calculations

Figure 2.18 shows the entire system and its subsystems, with the rectifier, buck converter and electrolyser shown in their entirety. a $200\text{ }\mu\text{F}$ capacitor in parallel, chosen empirically to provide sufficient smoothing and to supply instantaneous current to the buck converter during switching transients. Positioned above the rectifier is the controller subsystem, for which a more detailed view is given in Figure 2.21. It takes in both the measured power of the electrolyser and the reference power in kW, and outputs a three-element vector of PWM signals for the buck converter switches. The buck converter uses ideal switches connected in accordance with the model defined in Section 2.2.2, along with 2.47 m inductors and diodes of the same type as those used in the rectifier. A smoothing capacitor is again placed in parallel for lower ripple and instantaneous current supply to the electrolyser.

The electrolyser equivalent circuit model defined in Figure 2.4 is the load of the system, with a diode, resistors and a capacitor using the values defined in Table 2.2.

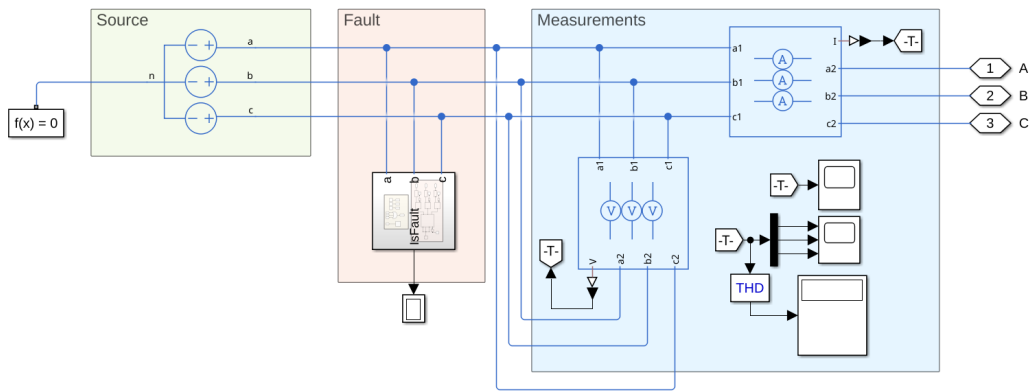


Figure 2.19: Source subsystem containing fault modelling, voltage and current monitoring and three phased voltage source running at $400V_{rms}$ as to simulate European mains power.

The source subsystem contains a three-phase voltage source representing a grid-conditioned supply. The source is configured with a phase-to-phase voltage of $400V_{rms}$, in accordance with the European low-voltage

grid standard. A purely resistive source impedance of $1\text{ m}\Omega$ per phase is included. Inductance has been omitted to avoid introducing additional poles into the plant model. Current and voltage monitors measure the values of all three phases and derive the total harmonic distortion.

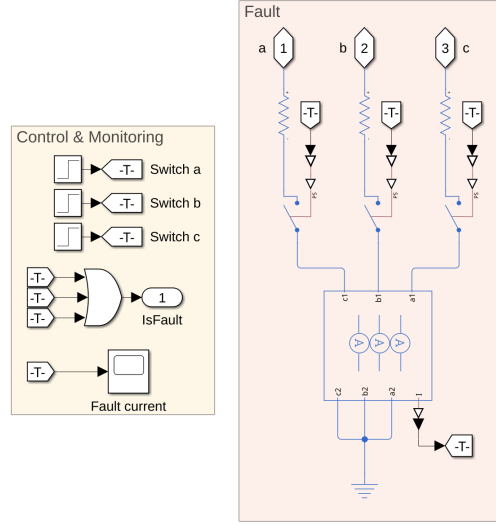


Figure 2.20: Fault model subsystem, modelled as phase-to-ground faults

The fault is modelled such that each phase is connected via a resistive element and a switch to ground. Step blocks allow for the fault to happen programmably for simulation purposes to test the effects of a fault using different control methods in a reproducible manner. Each resistive element in the fault sub circuit is chosen to be $0.1\text{ m}\Omega$ as to model it as a worst case, bolted fault.

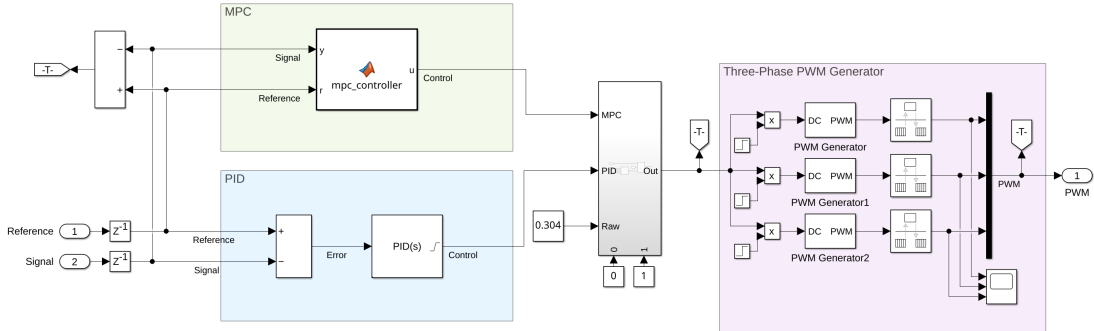


Figure 2.21: Controller subsystem, containing PID, MPC and direct feed forward duty cycle, as well as three-phase PWM generator system used for driving interleaved buck converter

The controller subsystem contains the three systems used for control over the electrolyser, namely the PID controller, the MPC controller and the uncontrolled duty cycle. Each method can be chosen using a binary counter during runtime, to allow for quick comparisons between said methods. The chosen controller then carries its signal into the interleaved PWM generator, which features three separate and isolated PWM generators running at identical frequencies of 1 kHz with a phase shift of 120° from each other. The PWM signals are then carried out of the subsystem to drive the switches of the buck converter. Due to the nature of signals being completely binary but driving a simulated physical switch, the outputs need rate limiting such that at no point does the signal change from 0 to 1 in a single time-step. But allows μs of ramp

up/down, which keeps the system running smoothly and not crashing from sudden ∞ changes in signal. The reference and signal ports are both delayed by a single time step as to prevent algebraic loops that occur when values depend directly on each other. Each PWM generator has a step block which can turn on or off the driving of the individual switches during runtime. This allows for recording the differences between using an interleaved buck and a single switch buck in real time, or simulating the event of a single switch failing open circuit, which is highly unlikely for MOS switches, but possible if high frequencies of large current break the copper wires, or control logic for one switch breaks.

The PID controller uses the built-in block from Simulink while the MPC controller uses a Matlab function as opposed to the ready-made Linear MPC block from Simulink. Both blocks are limited to an output in the range of $0 - 1$ as to stay within the limits of the PWM generators

Chapter 3

Controllers

3.1 Control methodology

As the system is built from buck converters that don't exactly function 1:1 with the math representing them, and electrolyzers which operation depend on many factors, a direct feed-forward control method is going to yield unfavourable results as there is no way to limit static offset or handle transients. Instead, a feed-back method is necessary to get accurate outputs and transients of the system. Feed-back controllers work by comparing an output of the system to a set reference, calculating what is known as "error". This error is then used to drive the control signal higher or lower, such that the error moves towards zero during a specific time-frame. There are plenty of feed-back controllers, however in this project only two are used, and compared to each other, the classic PID controller and the modern MPC controller.

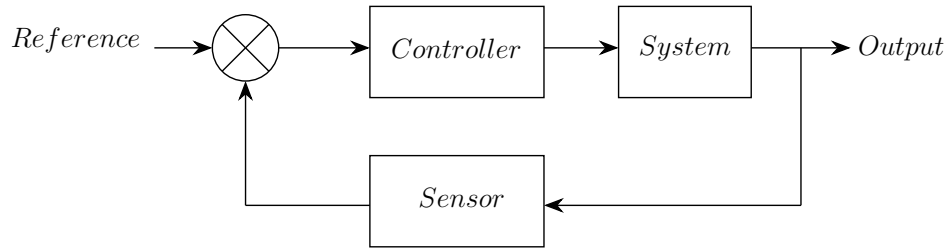


Figure 3.1: The general topology of a feedback-control system. Showing the movement of control data as sensor measurements of system variables are compared to a target reference to get an error which is interpreted by the controller such that it can change select variables of the system to bring about another output. This loop is the fundamental of feedback control.

This project aims to control the power draw of the electrolyser, and as such, the sensor and the feed-back signal will be gathered using the voltage and current measurements of the line entering the electrolyser, computing the power draw from said values and comparing them with a power draw target of 100 kW to get the error for the controller, this is the case for both controllers used in the project. The reason for controlling the power is that while current is the conventional controlled variable, controlling power allows the control of cost of the electrolyser, as well as possible temperature control by limiting target power.

3.1.1 PID

The most commonly used control algorithm is undoubtedly the PID (Product, Integral, Derivative) controller, which works by passing the error through three different stages in parallel and using the sum of those as the control signal. Different configurations exist, and not always are all three stages used, depending on the decisions made by the design engineer.

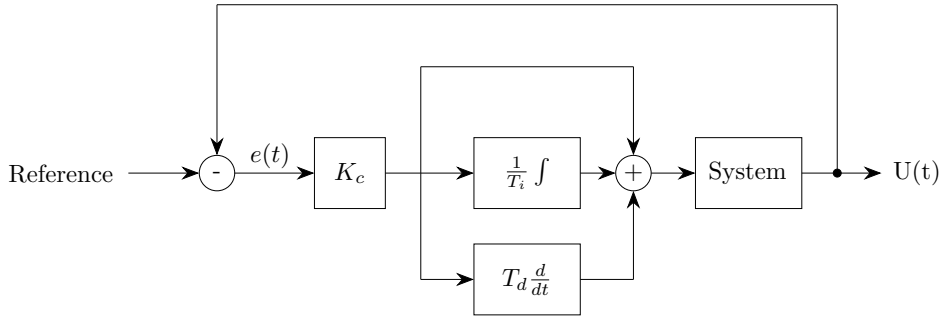


Figure 3.2: Topology of a PID controller with terms P, I and D shown as paths of the error signal $e(t)$ which process the data by multiplying, integrating and differentiating it and using the sum of said terms to drive the system, with the output of the system fed back to be compared to the reference

$$u(t) = K_c \left(e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau + T_d \frac{de(t)}{dt} \right) \quad (3.1)$$

Product

The product stage sets a general gain of the system. For many systems, the plausible values of the control signal is not comparable to the range of the actual output (i.e. 0%-100% for a motor, with angular speeds ranging from 0 to 4000 RPM). For this reason, the control signal must be scaled. For a linear system with no transient period, this would be all that is required to control a system, however, all systems feature varying different lengths of transient periods where the output settles after getting a new control signal, and thus, Integral and Derivative are needed.

Integral

The integral part measures how long error has been present in the system. The longer an error stays, the larger this part becomes and thus moving the control signal to correct for this. This is useful as it allows for much faster reaction times when an error has persisted for a short while. It also eliminates static offset, as small amounts of error built up gives rise to an increase in control signal amplitude to correct for said error. Setting this part high will result in overshoot. In some cases, small amounts of overshoot is tolerable if it allows for the system to settle faster, but for most systems, large amounts of overshoot is highly non preferable. Further increasing the integral part will result in oscillations of the control signal, where the error built up on one side of the reference point is enough to send the control signal to the other side repeatedly until the controller either settles or the oscillations become too large and the system breaks down.

Derivative

The derivative part aims to keep the slope of the error at zero, if the slope starts to get high, this part will increase the control signal amplitude accordingly to get ahead of any quick spike on the output. This part also helps reduce overshoot, as the overshoot can be seen as a high error slope which has to be corrected.

For the PID used in this project the values were tuned by hand from trial and error such that the system settled fast and contributed as little noise as possible. Most importantly was for the controller to be able to eliminate static error. The chosen values are found in table 3.1. These values include both the standard values of the PID (K_c , T_i , T_d) the wind-up coefficient, and the sample time of the controller, which was designed to run at the same rate as the simulation itself at $5\mu s$. Such as to quickly adjust to any changes in the system.

Variable	Value
K_c	0.5
T_i	0.005
T_d	0
T_b	2
T_s	$5\mu s$

Table 3.1: Values of the PID controller used for simulations

Since PID controllers only have knowledge of the instantaneous error-value, -slope and -area, it's possible for the control signal to become very large if error is not corrected in a time-frame excepted by the variables of the PID, i.e. if the actual slope of the signal is slower than that which is set by the values, or a target is unreachable due to physical limitations of the system. This causes what is known as wind-up, which is an ever increasing or decreasing control signal that moves beyond the range understood by the actuator. The consequence of wind-up is that, if the control signal has reached far beyond the maximum level accepted by the actuator, it will take some time for the signal to fall back to to a valid range of values again. This manifests as a 100% or 0% control value forcing the actuator to its maximum or minimum for an extended period of time. To fix this there are a few anti-wind-up methods. The most crude is the clamping method, where the integrator is frozen in the case that the controller is saturated and the error is at a value such that further increase in control signal would be needed, This method can still cause overshoot as the integral may still be very large at the point of freezing, and thus, needs time to correct for the integral error as well.

In this project, back calculation was used instead, as its better for very dynamic and fast systems. It works by including an additional term penalising control signals moving beyond maximum or minimum control values by adding an additional term $T_b(u_{sat} - u)$ in in equation 3.1 such that the full equation becomes

$$u(t) = K_c \left(e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau + T_d \frac{de(t)}{dt} + T_b(u_{sat} - u) \right) \quad (3.2)$$

Where T_b is the anti-windup coefficient, and $u_{sat} = \text{sat}(u, u_{max}, u_{min})$. As the control signal becomes larger than the Max or min values, the difference between u and u_{sat} increase to counteract the integrator and keep it in check such that once it is again possible to control the plant with values within the range of the actuator, it won't react sluggishly, but instead quickly correct the error. [3]. For this project the anti-wind-up coefficient of $T_b = 2$ was chosen empirically such that back calculations didn't act too aggressively and cause instability.

3.1.2 MPC

The model predictive control (MPC) is a control method that uses a mathematical model of the system itself to optimize for future output values and shapes the control signal to most efficiently bring the error to zero and apply only the control signal for the first time-step. Contrary to the PID controller, the MPC does

not simply measure the error and perform a calculation to yield an output, but creates, for each time-step, a quadratic optimisation problem where future output values are predicted and control signals are derived. There are two main versions of the controller, the linear and non-linear MPC. In this paper only the linear controller will be used and referred to simply as the MPC.

The MPC penalises both error and control slope meaning the optimisation problem needs to minimise both the error (Q) and the rate of change of the control signal (R). The MPC also does not predict infinitely far into the future as it would be computationally infeasible, it instead features a prediction window (N_p) and a control window (N_c). The prediction window states for how many time-steps the output value of the plant (x) are estimated using the given control signals at the previous time-step (u). The control window is for how many time steps the control signal is calculated. In the case that the prediction window is larger than the control window, the rest of the predictions will assume that time-step uses the same control signal as the last one calculated, so for $N_p = 10$ and $N_c = 3$, the predictions 4 through 10 will assume a control signal of $u_{k-i} = u_{k-3}$ [10]

General workings

The MPC takes in a linear model of the system that is to be controlled, This equation is the basis for the MPC controller (Eq. 3.3, 3.4). The first value x_k is the actual measured states of the system, this can be a single variable like voltage output of a buck, or multiple sensor values like the voltage, temperature, frequency etc. of the plant.

$$x_{k+1} = Ax_k + Bu_k \quad (3.3)$$

$$y_k = Cx_k \quad (3.4)$$

This vector is then used to calculate the following estimated states of the system with control signals $u_k \dots u_{k+N-1}$ applied (Eq. 3.5)

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k \\ x_{k+2} &= Ax_{k+1} + Bu_{k+1} \end{aligned} \quad (3.5)$$

This can be congregated such that each state across the different time-steps are a separate vector where X & U are all the states and control signals and F & G are prediction matrices defined by the state space matrices (Eq. 3.6, 3.7).

$$X = Fx_k + GU \quad (3.6)$$

$$\begin{aligned} U &= \begin{bmatrix} u_k & u_{k+1} & \dots & u_{k+N-1} \end{bmatrix} \\ X &= \begin{bmatrix} x_k & x_{k+1} & \dots & x_{k+N-1} \end{bmatrix} \end{aligned} \quad (3.7)$$

In order to actually calculate the future control signal values, the cost function needs to be defined. The function, just like with PID, calculates an error by comparing the sensor reading x_i with the reference value x_{ref} . This error is multiplied by the weight Q which defined the importance of the error, the higher the Q value, the more the cost function will favour eliminating error compared to keeping the control signal slope low. For the slope, the slope is multiplied by the weight R which, just like Q , decides the importance/cost of control signal slope. A higher R will yield a more smooth controller and mitigate overshoot.

$$J = \sum_{i=1}^N ((x_i - x_{ref})^T Q \cdot (x_i - x_{ref}) + i_{i-1}^T R \cdot u_{i-1}) \quad (3.8)$$

$$J = \frac{1}{2}U^T H U + f^T U \quad (3.9)$$

Hard Constraints

A big part of MPC controllers is the option to have constraints built into the controller. This allows defining limits of different parts of the cost function. Most controllers set limits for the control signal of actuators as to not overload them or cause dangerous scenarios of motors spinning too fast, however it's entirely possible to also set constraints for the control signal slope Δu , such that even if a large slope is the cheapest solution according to the controller, it's still hard limited below a certain threshold.

$$u_{min} \leq u_k \leq u_{max} \quad (3.10)$$

Constraints don't have to be static and are also able to be equations as long as they're defined by controller variables like states, control signals and such. For linear MPC controllers, any constraints must be linear in nature as well. For this, a non-linear MPC is required.

Hard constraints are included in the optimisation problem but are not present in the cost function, solvers are able to use the constraints to iterate through possible solutions until the solution minimises the cost and satisfies all hard constraints.

Soft constraints

Constraints don't have to be hard limits, soft constraints can be used to heavily penalise passing certain thresholds, this could be used to smoothly decrease motor acceleration as it passes top speed such that it doesn't stop increasing instantly, but slows the acceleration heavily to keep from passing the threshold more than it already has. This is also useful for controlling the temperature of a system where it's fine to deviate a bit within reason, but passing certain thresholds would be expensive and only possible if absolutely necessary.

Soft constraints are included in the cost function as additional terms, for a state variable penalised for crossing a threshold the function

$$s = \max(0, T - T_{max}) \quad (3.11)$$

This can then be appended to the cost function with a large multiplier to make the cost of crossing beyond the threshold very expensive

$$J = J_{original} + w \cdot s^2 \quad (3.12)$$

This function is used by the MPC to calculate the production matrices.

```

1  for i = 1:Np
2      F(i,:) = C_aug*(A_aug^i);
3      for j = 1:Nc
4          if i >= j
5              Phi(i,j) = C_aug*(A_aug^(i-j))*B_aug;
6          end
7      end
8  end

```

With only the system transfer function it's clear that the inaccuracy of the model regarding steady state operating will translate to the controller itself, and steady state error will be present regardless of how high the weights against said error is. For this reason, the MPC needs to also know of the integral of the system,

the cumulative error recorded by the sensor. This is done by augmenting the state space system derived from the transfer function such that the A matrix has a row dedicated to this error which the MPC can penalise heavily.

```

1  A_aug = [A zeros(nx,1); -C 1];
2  B_aug = [B;0];
3  C_aug = [C 0];

```

Adding this little change allowed the MPC to settle quickly and accurately, even despite the large non-linearities, since the transient periods themselves were linear.

The final values used for the MPC are seen in table 3.2, with the sample time being equal to that of the simulation step-size itself, as was the case with the PID controller, a prediction horizon of 100 and a control horizon of 10. The prediction horizon was chosen as quite large due to the high sample rate of the controller such that it would attempt to predict $500 \mu s$ into the future. To limit computation the control horizon was chosen as a tenth of the prediction horizon, such that it did in fact predict control manoeuvres, but didn't have to solve a cost-function a hundred times every $5 \mu s$. The large output weight and low input weight was chosen empirically such as to minimize static error of the system stemming from the controller deeming it cheaper to stay at zero slope than to correct for the remaining error. At higher Q/R ratios, the system would penalise slope too little and additional overshoot noise would occur.

Table 3.2: MPC Values

Name	Symbol	Value
Sample time	T_s	$5 \mu s$
Prediction horizon	N_p	100
Control horizon	N_c	10
Output Weight (penalise error)	Q	500
Input weight (penalise slope)	R	1

The current implementation does not include constraints and thus require no iterative solver, as the optimal solution at any given time is not unachievable due to set boundaries.

Chapter 4

Results & Discussion

This chapter will go through the output of the full system subjected to a range of different scenarios to record how well the controllers are at suppressing grid noise, the noise of different control methods, interleaved buck failure, grid fault scenarios and stepped target values. These results will then be discussed and explained, Table 4.1 contains the full list of scenarios applied to the system in this chapter. The expectation is that MPC outperforms PID in disturbance response and harmonic rejection, while both demonstrate satisfactory noise suppression and fault handling

Scenarios
Stepped Target Values
Switch Failure
Grid Fault
Harmonic Distortion

Table 4.1: List of simulation scenarios for PEM system

4.1 Noise / Ripple Suppression

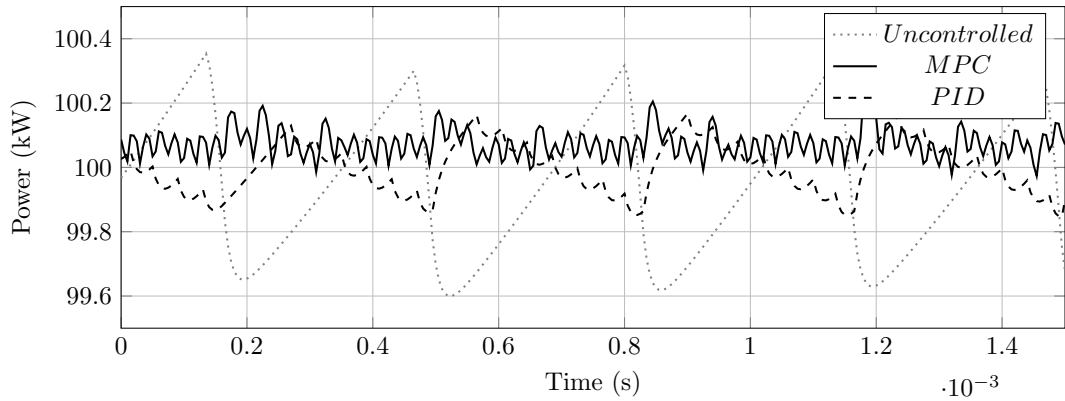


Figure 4.1: Electrolyser power draw comparison between PID, MPC and uncontrolled system with $D = 0.68$

Figure 4.1 shows a comparison between the noise of the two controlled systems with a reference to the non-controlled system. It's clear that the uncontrolled system has much larger power ripple compared with the controlled versions, Figure 4.2 shows the frequency content of 2 seconds of recorded power draw, estimated

using Welch's method with a Hann window of size 65536 and a sampling frequency of 200 kHz , expressed in dB relative to $1\frac{\text{W}^2}{\text{Hz}}$. In total, three frequencies dominate the raw signal, 300 Hz , 1 kHz and 3 kHz .

The 300 Hz spike matches that of the rectified three-phase source seen in figure 2.11. Two harmonics of this frequency are seen at 600 Hz and 900 Hz . The spikes at 1 kHz and 3 kHz are most likely results of the switching power supply, as each phase is switched at 1 kHz , but combined result in an effective frequency of 3 kHz since their phases are spaced apart equally. The remaining frequency spikes of the uncontrolled system are clearly harmonics of these three dominating frequencies.

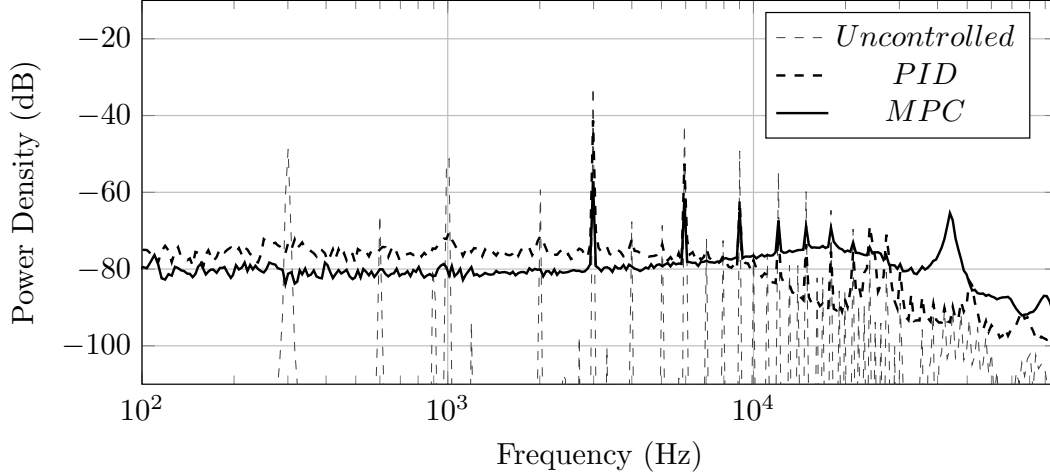


Figure 4.2: Power Spectral Density of PID and MPC simulated over the course of 2s of steady state operation

Figure 4.2 shows that both manage to suppress the large spikes at 300 Hz , 1 kHz and 3 kHz with varying degrees of success. The MPC shows better suppression of all spikes as well as a lower noise floor than that of the PID for frequencies below 6 kHz . However for frequencies above this point the MPC seems to induce more noise with which can also be seen in figure 4.1 where the MPC has more high frequency components than the PID. Both controllers result in lower ripple overall, but at the cost of a generally higher noise floor of about -80 dB .

4.2 Harmonics

The harmonic distortion test was done using the source block from Simulink and encoding different levels of harmonics inside the three-phase output. The frequencies chosen were and their amplitudes are shown in table 4.2.

Harmonic	Frequency	Amplitude
1 st	50 Hz	100%
2 nd	100 Hz	10%
3 rd	150 Hz	30%
4 th	200 Hz	1%
5 th	250 Hz	20%
6 th	300 Hz	1%
7 th	350 Hz	10%

Table 4.2: Harmonics introduced to the system

The THD (Total Harmonic Distortion) of the values in table 4.2 is 38.755% which is an extremely high level, thus, lower values distortion was also tried against the system by introducing a multiplier to only the harmonics and not the fundamental. This multiplier would be set to five different levels which resulted in the THD values seen in table 4.3. The resulting waveform of these levels of THD can also be seen in figure 4.3. The values 0, 0.25, 0.5, 0.75 and 1 were selected to test the controller at different extreme cases of THD.

Mult	THD
1.00	38.755%
0.75	29.066%
0.50	19.377%
0.25	9.688%
0.00	0%

Table 4.3: Multiplier values for the harmonics of the system and their corresponding calculated THD values

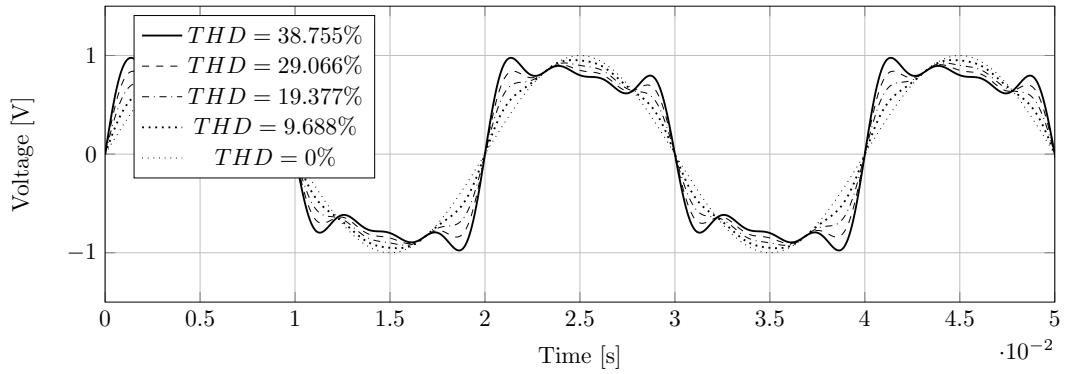


Figure 4.3: Example of the effect of large levels of harmonic distortion on signal integrity

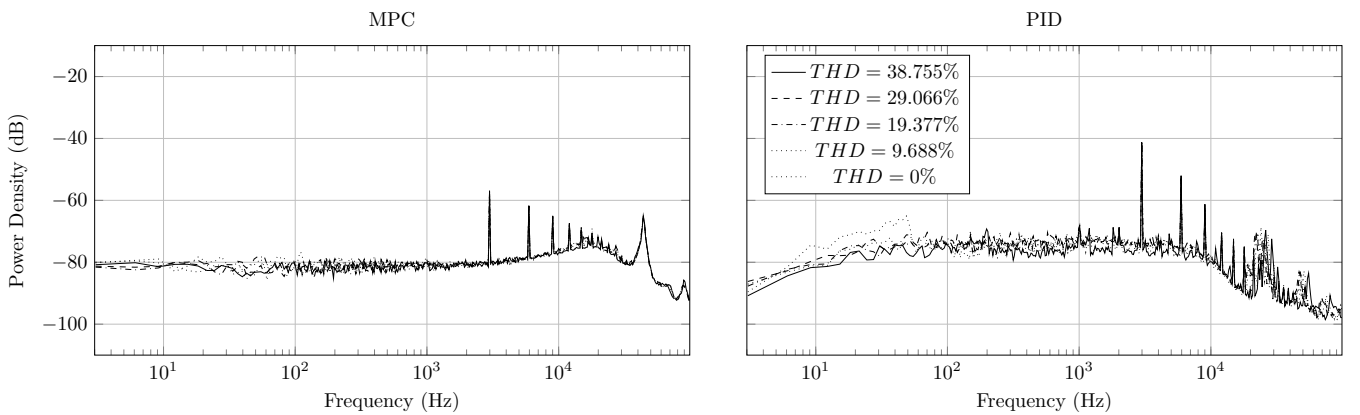


Figure 4.4: Comparison of time domain interpretations at different levels of Total Harmonic Distortion controlled using the MPC setup

Table 4.4: Electrolyser output power ripple under harmonic distortion

Method	THD	Error. (kW)	RMS ripple (%)	P-P ripple (%)
MPC	0%	0.072	0.045	0.342
MPC	9.688%	0.072	0.045	0.34
MPC	19.377%	0.072	0.045	0.338
MPC	29.066%	0.072	0.045	0.344
MPC	38.755%	0.072	0.045	0.336
PID	0%	0.0	0.078	0.359
PID	9.688%	0.0	0.078	0.362
PID	19.377%	0.0	0.077	0.361
PID	29.066%	0.0	0.077	0.356
PID	38.755%	0.0	0.08	0.354

Running the system for both PID and MPC it's immediately clear to see from table ?? that the PID achieves lower ripple across all THD levels than the MPC in all aspects. It showed lower ripple for all levels of THD and no static error at all. And with a difference between the lowest and highest levels of RMS ripple of just 2.78%, the distortion has a negligible effect on the output of the system. Looking to figure 4.4 it can however be seen that the amplitude of the frequencies of $200Hz$ and up (Remember, the system is rectified, doubling the effective frequency of the source) do in fact increase, but only marginally. The MPC however, sees a difference between highest and lowest RMS ripple of 6.65%, a 139% increase compared to that of the PID. And again, looking to figure 4.4 it's clear to see the large differences in frequency content in the lower frequencies around $200Hz$ and up. Interestingly though, the MPC seems to have a lower amplitude in all frequencies below $2kHz$ for the highest THD level. This might be worth looking more into, as the level does increase reliably for the other levels, but jumps down sharply for the 38.755% case.

The best performance in terms of handling high levels of harmonic distortion seems to come from the PID, however, as most harmonic distortion is in the range of 0-5% THD, the low differences in output noise in the high ranges prove that both handle the input distortion rather well.

4.3 Switch Failure

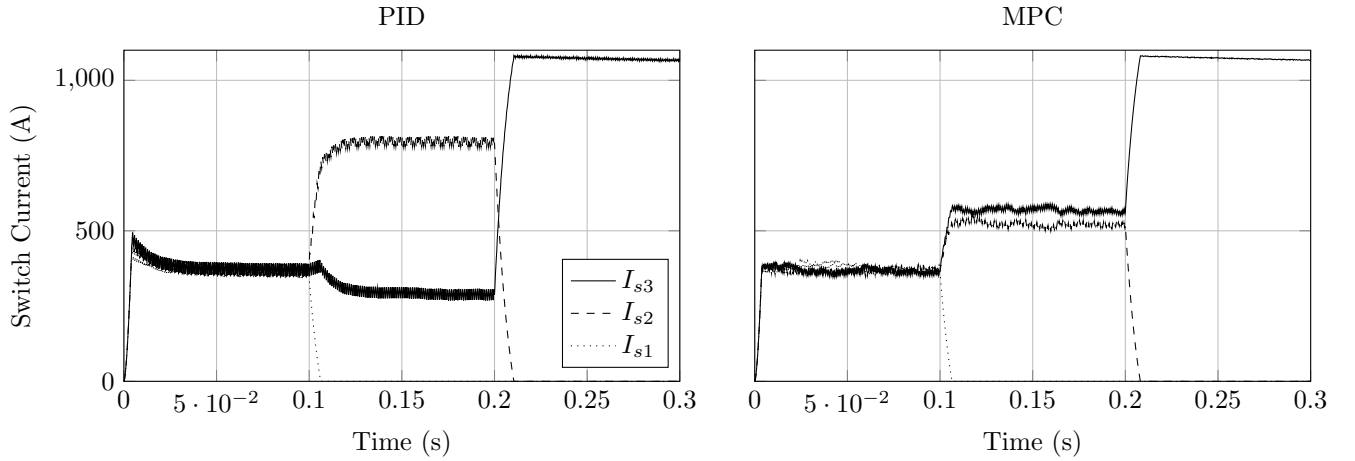


Figure 4.5: Interleaved buck switch open circuit failure, switch 1 fails at $t=$ and switch 2 fails at $t=$, current is concentrated in the switches still operational and control keeps power at target value of $100kW$

Figure 4.5 shows the current measured as running through each separate inductor in the buck converter, at times $1ms$ and $2ms$ switch 1 and switch 2 fails as open circuit and stops conducting current. While MOSFETs aren't known to fail as open circuit, it's still interesting to review the case that a single switch goes offline, this might happen due to wires breaking or system glitches of the PWM generators causing one or more switches to stop conducting current. It is here assumed that the switches of the system are rated to $600A$, and as such, will fail if a higher current flows through the phase.

For the MPC, the sharing of the load isn't quite as equal as for the PID at any given time, with slight variation over the course of the first $0.1s$. However taking the mean of the MPC the switches do share the load fairly equally as seen in the first column of figure 4.5. After the first switch fails the two remaining don't share quite the same current, with one running at a mean of $522.2A$ and the other at $569.5A$, though both are relatively stable and neither switch peaks above $600A$, although switch I_{s3} comes close at $595A$. After the second switch fails the remaining end up having to supply $1073.5A$, surpassing the current rating of the switch by $1.79x$.

Phase	Pre-fault (A)	Post-fail 1 (A)	Post-fail 2 (A)
Is1	369.5	0.0	0.0
Is2	358.6	522.2	0.0
Is3	360.0	569.5	1073.5
Total	1088.1	1091.7	1073.5

Table 4.5: Switch failure current redistribution (MPC)

The PID starts out by showing a large overshoot as will also be mentioned in 4.4. Peaking at $502A$ and settling to the same 360 as the MPC. The biggest most obvious difference is in how poorly the PID case handles equalizing the phases once the first switch fails, immediately passing the rating of switch 2 by increasing the current through it to just under $800A$ while switch 3 runs at just under $300A$. The reason for this is hard to discern, as the controller doesn't have any control over the share of the current, only the duty cycle of all switches. However one possible reason is that small differences in current level of switches

at the time of failure could cause more current to pass through switch 2. And since no control is applied to mitigate inequality between the switches. This difference only grows larger as the current increases to compensate for the missing switch. Whether or not it's a control issue, a modelling issue or something else would have to be investigated further.

Phase	Pre-fault (A)	Post-fail 1 (A)	Post-fail 2 (A)
Is1	372.2	0.0	0.0
Is2	365.6	793.0	0.0
Is3	375.5	297.6	1072.4
Total	1113.3	1090.6	1072.5

Table 4.6: Switch failure current redistribution (PID)

4.4 Stepped Response

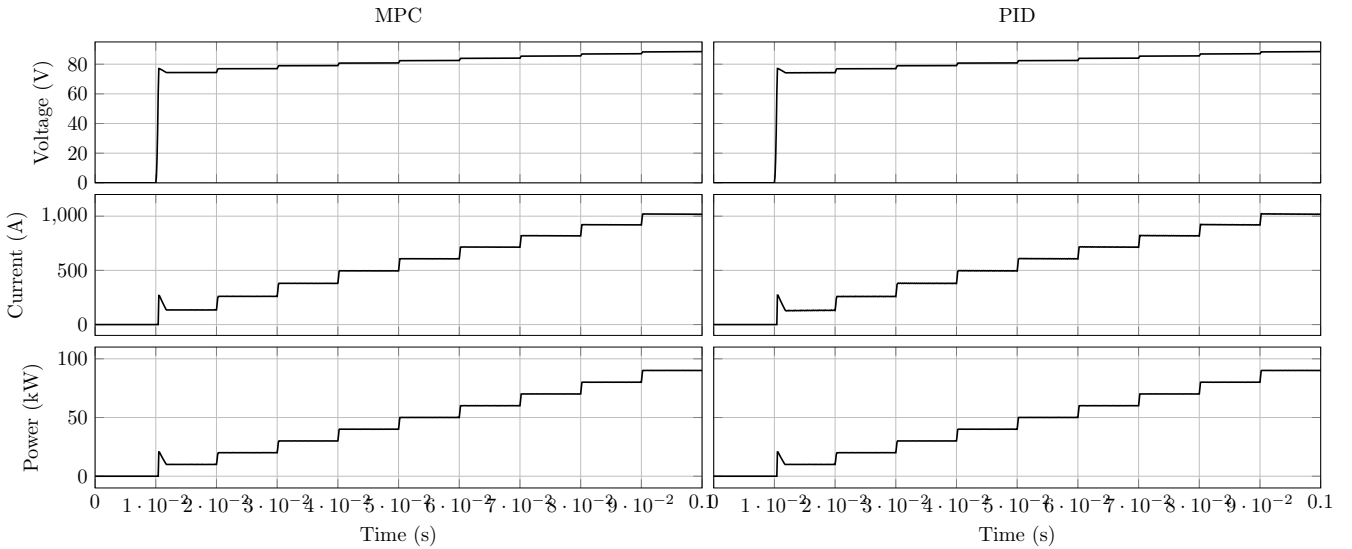


Figure 4.6: Stepped response of PID and MPC with $10kW$ steps at $10ms$ intervals until the system reaches $100kW$

When stepped in equal increments of target power, a few things can be seen, first and foremost both PID and MPC controller are fast to settle at most of the steps, however the first step from $0 - 10kW$ shows a larger overshoot from both controllers in the power of the system. This is most likely due to the fact that power is the product of both voltage and current, and as such, as the system only conducts after roughly $75V$ as determined earlier in figure 2.17, a delay between increased control signal and power output at start-up can be seen, resulting in wind-up and overshoot. This is cooperated by the fact that both controllers witness this effect, and that as soon as the voltage is above $75V$, such as for the rest of the steps of the system, the overshoot is no longer present for either controller.

At a macro-scale, both controllers step in very identical manners, with very low variance in power as was also determined in section 4.1. There are however some dynamics differences between the two on the short-time scale. as such both systems. In order to properly compare the step-response of the two controllers without the system start wind-up interfering, a single step to $100kW$ was recorded for both controllers, with

each being pre-conditioned by starting at $0.001kW$ such that the system was already conducting and the voltage levels already beyond the threshold of electrolyser (Fig. 4.7).

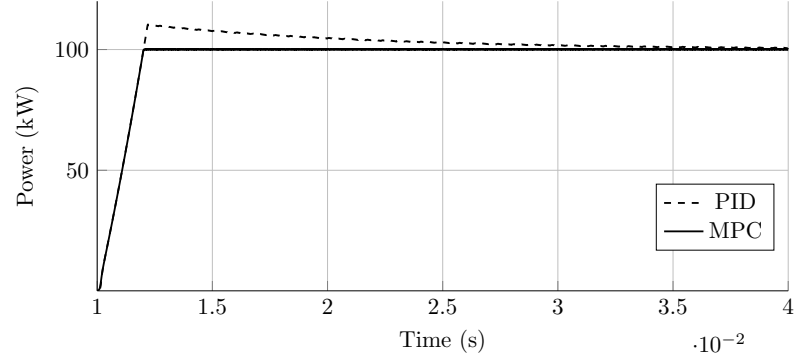


Figure 4.7: Step response of controlled system comparison between MPC and PID tuning

Table 4.7: Step response metrics for MPC and PID controllers

Metric	MPC	PID
Rise time (ms)	1.575	1.575
Settling time (ms)	11.975	29.115
Overshoot (%)	0.251	10.36
Steady-state err (kW)	0.072	0.0
Control effort (kW/ms)	14.212	12.428

Figure 4.7 shows a comparison between the transients of the system when controlled by either the MPC or PID controller. While the two are very similar in slope, the PID is seen overshooting the target ever so slightly before settling over the course of $20ms$.

4.5 Fault Scenario

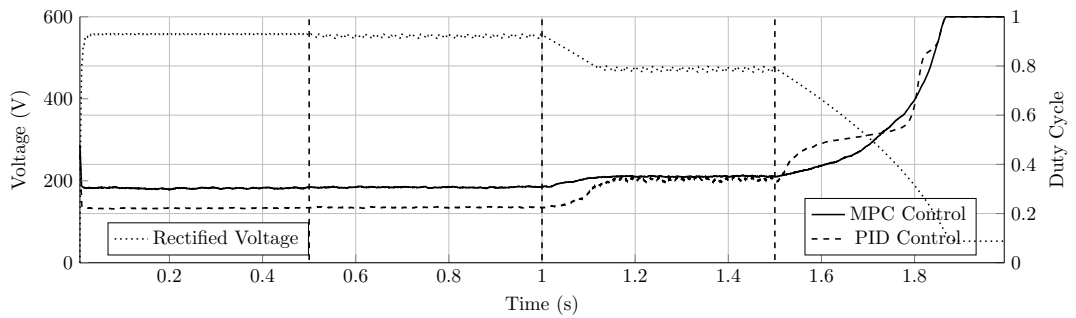


Figure 4.8: Control reaction and rectifier voltage of system under different phase-to-ground fault scenarios, the rectifier voltage is shown with values to the left side, the control signal of the PID and MPC are shown with values to the right side

As seen in figure 2.20 the grid fault scenario consists of three phase to ground connections with an impedance of $0.1m\Omega$ to simulate a bolted connection.

The scenario has three stages, the first stage at $500ms$ switches the first fault on, the second stage at $1s$ switches the second fault on, at $1.5s$ all phases are grounded through their inductances, causing very little current to run to the plant. Since the control signal the PID and MPC mostly vary from 0% to 100%, a running average of the control signals are presented here in place of the raw control signals such as to give the reader more clarity. The running average used a window size of 3000 samples with 1 sample of overlap ($15ms$), values were chosen as to limit noise but not cause the signals to appear slower to react to changes than they are.

The results in figure 4.8 show that for a single phase to ground fault, the drop in voltage after the rectifier isn't very severe, and the controller barely has to increase the control signal to compensate for the loss of current through ground. At the second fault, the voltage drops by about 13% which the controller is seen compensating for by increasing the duty cycle accordingly from around 30% to 35%. The last test was the quite unrealistic case where every phase was grounded. Here, the controller attempts to significantly increase the duty cycle to compensate before it reaches saturation, at which point the electrolyser output power to zero as can be seen in figure 4.9.

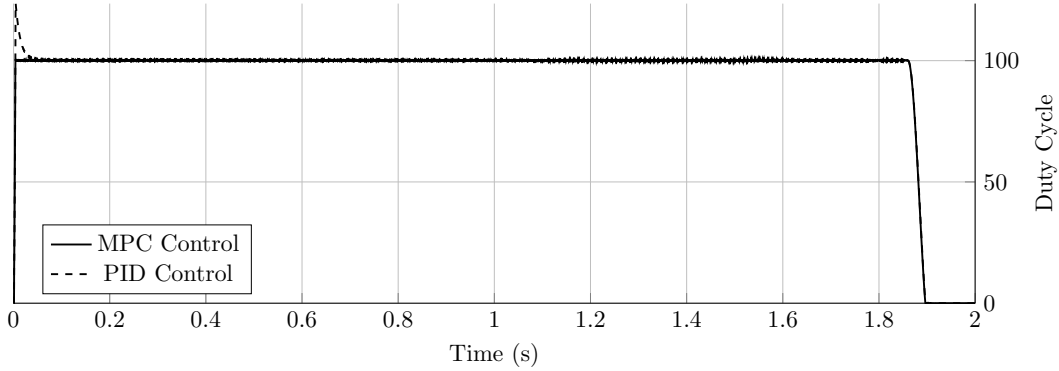


Figure 4.9: Electrolyser Power draw during phase-to-ground fault scenario

Chapter 5

Conclusion & Future Work

The project set out to create an equivalent model of a PEM electrolyser and the power delivery circuitry in Simulink, and apply different control methods to monitor the differences between the methods, as well as determine whether the model predictive controller is viable as a method to use with proton exchange membrane electrolyzers. The system was run through different scenarios such as grid fault, switch failure, extreme harmonic distortion and stepped control targets.

The complete system was completed with three phased source integration, rectification interleaved buck converter for stepping down the voltage and the electrolyser equivalent model derived from polarisation curve models as well as a complex example setup [21]. A linearised transfer function capturing the dynamics of the PEM electrolyser was also derived and compared to the example system as well as the circuit equivalent and used to drive the MPC controller as effectively as possible due to the input of the function relating a given duty cycle to a power output. The model was made to fit the system at a given duty cycle of $d = 0.1766$, and although the system itself draws 100kW at $d = 0.301$ it still remained stable throughout the full range of duty cycles.

The MPC controller itself was coded from scratch using the Matlab scripting language instead of using the built-in MPC block. The reason for this was that the block from Mathworks® wasn't able to remain stable for any time-step or any configuration. The MPC script proved to not only be more robust than the built-in system, but also enabled fast change of the entire system, such as allowing the addition of a term penalising static error as explained in section 3.1.2.

The general quantitative finding showed strengths and weaknesses for both controllers. Most notably the both PID and MPC has equal rise-times and on a macro-scale, they operate very similar. However the PID shows significantly higher levels of overshoot (0.251% vs 10.36%), slower settling time (11.975 *ms* vs 29.115 *ms*), worse suppression of switching dynamics and worse low frequency noise floor 4.2. The MPC however does feature static error, which the PID does not. Due to the way the MPC is a cost-function, lower cost of error results in potentially higher overshoot as the controller attempts to correct. Both controllers showed strong rejection of input harmonic distortion, however the MPC did perform better than the PID in terms of difference between zero and high levels of distortion, with a barely any difference for 1 *kHz* and up, and very minor differences below 1 *kHz*, while the PID had a bit more variance below 100 *Hz* and above 10 *kHz*, most notably at around 11 *kHz*. The two controllers handled the switch failure very differently, with the PID case handling sharing of current better than the MPC before any switch failed, but much worse after the first failure where one switch conducts 793.0 *A* and the other 297.6 *A*. The MPC on the other hand handles the sharing of current better after the first failure. As discussed in the results, the cause of this is currently unknown, however speculations lean towards the anomaly being less the fault of the controller itself, and more to do with circumstances surrounding the simulation such as the ratio between the chosen switching frequency and the sampling frequency of the system itself. Figuring it out would require additional

troubleshooting and research. The source used in the project is also a simple three-phase source. Which does not feature variable output power such as renewables have. Using a mathematical model of renewable generation, it would be interesting to test the ability to use the MPC to forecast generation and plan hydrogen output according to said mathematical model and examine the implications of such forecasting. The both controllers handled grid faults equally well, since the system rectifies the three-phase $400V_{rms}$ source into a single DC line and then only uses around $95V$ there is a lot of headroom for receiving a lower voltage and still being able to run at the $100kW$ without much issue, at least setup used in this project with $1m\Omega$ source impedance and $100\mu\Omega$ phase-to-ground impedance. Both controllers kept the output power stable right until the rectifier voltage went below $100V$ and the output began to drop, ending up below the threshold of the electrolyser determined by the polarisation curve and halting production completely.

Some limitations caused assumptions and simplification on the system to be made. The biggest were the lack of temperature dynamics and concentration voltages as well as system pressure. All of these limitations are caused by the decision to use of equivalent circuit model instead of modelling the electrolyser as a controlled voltage source or impedance. For temperature dynamics the diode forward voltage would need to be variable and calculated separately and a temperature estimation term would have to be included, the same applies for concentration the voltage, as it's not possible to build an equivalent circuit model from passive components that allows for an exponential rise in impedance past a certain voltage threshold. The frequencies of the controllers compared to the PWM signal was also quite curious. The reason for the low frequency of the PWM was that as the PWM frequency and sample rate of the entire simulation got closer, the resolution of the PWM signal got smaller since there were less time-steps to represent any given duty cycle. This got especially bad once the interleaving buck converter was introduced and caused major issues with stability and signal error and thus it was decided that the two frequencies remain far apart. However in decreasing simulation sample time to have higher PWM resolution the time to simulate and the amount of data points made simulations incredibly slow and memory hungry to the point where simulations spanning a few seconds would crash Simulink. So as a compromise a low PWM frequency was accepted.

In future works it would be worth looking into creating a robust framework for using a controlled impedance or voltage source to create an electrolyser model which is computationally efficient, allows for temperature, pressure and hydrogen generation estimation, and includes feedback of these variables such that very slow dynamics can be modelled as well. This would also allow for using said variables in the MPC, unlocking the full potential of its multi-variable design and allowing it to keep track of multiple parts of the system to make the control decisions smarter and testing the use of a single central controller versus multiple dedicated controllers.

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